

Combinatorial Coefficient Calculus

**Stirling, Bell, Eulerian, and Bernoulli–Euler Families;
Bell-Polynomial and Faà di Bruno Calculus;
Lagrange–Bürmann and Multivariate Good Inversion**

A self-contained synthesis with proofs

September 2026

Abstract

This monograph develops a unified calculus for the coefficients that occur when one changes polynomial bases, partitions labelled sets, differentiates or composes functions, and reverses formal or analytic power series. Its principal objects are the signed and unsigned Stirling numbers of the first kind, the Stirling numbers of the second kind, Lah numbers, Bell numbers, exponential and ordinary Bell polynomials, Eulerian numbers of types A and B , second-order Eulerian numbers, Faà di Bruno's formula, and the Lagrange inversion and reversion theorems.

The treatment is deliberately proof-oriented. Definitions are connected by bijections and change-of-basis arguments; generating functions are derived rather than merely recorded; recurrences, convolutions, congruences, determinant formulae, probabilistic interpretations, special-function identities, and asymptotic formulae are proved from a small collection of reusable principles. Applications include finite differences, differential-operator normal ordering, moments and cumulants, symmetric polynomials, cycle indices, Hermite polynomials, polynomial sequences of binomial type, negative polylogarithms, Bernoulli-number identities, Poisson moments, Catalan and Fuss–Catalan series, the Lambert W function, and Laplace-type asymptotic integrals.

The five source drafts are merged at theorem level rather than concatenated. Their additional strands—incidence algebras, Möbius inversion on the partition lattice, Sheffer–Riordan calculus, Fréchet derivatives, inverse-map recursions, and Good's multivariate inversion formula—are developed in full. The Wikipedia and MathWorld dossiers contribute a further unified treatment of Bernoulli, Euler, Genocchi, and Cauchy–Gregory families; Pochhammer symbols, digamma and polygamma functions, Barnes' G -function; complementary Bell numbers; Catalan–Narayana and associahedral refinements; Darboux formulas and singularity transfer; exact coefficient algorithms; and complete worked inverse-series examples including Kepler's equation.

The sources use the letter B for Bell numbers, Bell polynomials, Bernoulli numbers, type- B Eulerian numbers, and binary-tree numbers. Here these are separated notationally. A final concordance identifies the contribution of each draft and each reference dossier, while a proof-audit ledger records normalized, corrected, analytic, and computational statements separately.

How to read this work

The chapters are arranged by mathematical dependence, not by the order of the source files. The two master ideas are:

1. a coefficient is often easiest to compute after changing to the basis in which the relevant operator is diagonal; and
2. an exponential generating function remembers the decomposition of a labelled object into components.

Stirling numbers implement the first idea. Bell polynomials implement the second. Eulerian numbers mediate between powers and shifted binomial coefficients. Faà di Bruno’s formula is the differential form of labelled composition, while Lagrange inversion is the coefficient form of implicit substitution.

A result stated for formal power series requires no convergence. When an analytic version is asserted, its domain hypotheses are written explicitly. Statements in the archive that report historical computations—for example, lists of Bell primes or verified exact modular periods—are distinguished from theorems and are accompanied by a certificate procedure rather than misrepresented as symbolic identities.

Editorial normalization

Three systematic repairs are used throughout.

1. Bell numbers are denoted B_n , complete exponential Bell polynomials by $\mathcal{B}_n$, partial Bell polynomials by $\mathcal{B}_{n,k}$, and Bernoulli numbers by β_n .
2. Type- B Eulerian numbers are indexed by descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift in one formula in the archive.
3. The archived Moser–Wyman display contains unnamed quantities B, P_i, Q_i . We give the exact saddle-point scheme that defines every correction coefficient recursively and state the leading expansion in closed form; no undefined placeholder is retained.

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Notation

For $n \in \mathbb{N}$, write

$$x^{\overline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\underline{n}} = x(x+1) \cdots (x+n-1),$$

with both expressions equal to 1 when $n = 0$. Coefficient extraction is denoted by $[z^n]F(z)$, and $\text{Res}_z F(z) dz$ means the coefficient of z^{-1} in a formal Laurent series. Our principal arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle,$$

where $s(n, k)$ is the signed first-kind Stirling number, $\begin{bmatrix} n \\ k \end{bmatrix}$ its unsigned version, $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ the second-kind Stirling number, and $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ the ordinary Eulerian number. Bell numbers are B_n ; exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$.

Unless stated otherwise, sums over an impossible index range are zero, and $0^0 = 1$ in coefficient formulae. Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,

$$[z^n]F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n]F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}]A(z)$. Then

$$[z^n]F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.2 (Formal change of variables for residues). Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,

$$\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + O(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$\hat{A}(z)\hat{B}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta x^{\underline{k}} = kx^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $x + 1^{\underline{k}} - x^{\underline{k}} = kx^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.3 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} x^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k x^{\underline{k}}$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n(x_1, \dots, x_n)$.

Theorem 1.4 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Corollary 1.5 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $x^{\overline{n}} = (-1)^n - x^{\underline{n}}$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $x^{\overline{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n=3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x=1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). For $n, k \geq 0$,

$$\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). For all $n, k \geq 0$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

The simple diagonals follow at once:

$$\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1, \quad \left\{ \begin{matrix} n \\ 1 \end{matrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{matrix} n \\ n-1 \end{matrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.6 (Alternative second-kind recurrences). For $1 \leq k < n$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{matrix} n \\ k+j-1 \end{matrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{matrix} n-j+1 \\ k \end{matrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{matrix} n \\ r \end{matrix} \right\} k^r$. Since $k^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\binom{-k}{j} = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.7 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = y^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{y^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x) \log(1-x)) C'_k(x).$$

The left side has generating function $x C'_k - k C_k$. Their equality reduces to

$$-(1-x) \log(1-x) C'_k(x) = k C_k(x),$$

which follows immediately from the displayed formula for C_k . □

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1							0	1						
1	0	1						1	0	1					
2	0	1	1					2	0	1	1				
3	0	2	3	1				3	0	1	3	1			
4	0	6	11	6	1			4	0	1	7	6	1		
5	0	24	50	35	10	1		5	0	1	15	25	10	1	
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n \}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $x^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Corollary 3.2 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} x - 1^{\underline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} n^{\underline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $x^{\underline{k}} = xx - 1^{\underline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Corollary 3.3 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\{n_k\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand x^n into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{N+1^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2}N + 1^2 + \frac{7}{3}N + 1^3 + \frac{6}{4}N + 1^4 + \frac{1}{5}N + 1^5,$$

which expands to the familiar quartic power-sum polynomial.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.4 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}X^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{r^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} X^{\underline{k}}$. $\square$

Theorem 3.5 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n .

Proof. The random variable Y_m^k counts ordered k -tuples of distinct fixed points. There are m^k candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}Y_m^k = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}Y_m^k = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. $\square$

3.4 Lah numbers

Definition 3.6. For $1 \leq k \leq n$, the unsigned Lah number is

$$L(n, k) = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $L(0, 0) = 1$ and zero otherwise outside the triangle.

Combinatorially, $L(n, k)$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.7 (Conversion between rising and falling factorials).

$$x^{\bar{n}} = \sum_{k=0}^n L(n, k) x^k, \quad (3.16)$$

$$x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\bar{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} L(n, j) L(j, k) = \delta_{nk}. \quad (3.18)$$

Moreover,

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} L(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k L(n, k) x^k \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} x^{\bar{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq L(n, k) \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.2 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} 1/x^{k+1}}. \quad (4.6)$$

Consequently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\begin{Bmatrix} n \\ r \end{Bmatrix}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ by $k!$.

4.3 The Stirling transform

Theorem 4.3 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$\widehat{G}(z) = \widehat{F}(e^z - 1), \quad \widehat{F}(z) = \widehat{G}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.3. For the generating function,

$$\widehat{G}(z) = \sum_k f_k \sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = \widehat{F}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $m^{\underline{k}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.4 (Stirling normal-ordering identities). *For $n \geq 0$,*

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = x D^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^{\underline{n}} = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} m^{\underline{k}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $m^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, 2 + 2 for the second diagonal and 4, 3 + 2, 2 + 2 + 2 for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m[x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ n \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ j \end{matrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx} x^{\bar{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X+c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.2 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j-k$ and lower index $j-n$:

$$\begin{Bmatrix} j-k \\ j-n \end{Bmatrix} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-m-n+k)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.3 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{\prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p=0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p=0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1+A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$x^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $x^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $x^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

6.2 Parity of second-kind numbers

Theorem 6.2 (Parity criterion). *Let*

$$z = n - \left\lceil \frac{k+1}{2} \right\rceil, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv \binom{z}{w} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\left\{ \begin{matrix} 2n \\ n \end{matrix} \right\}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\binom{n-k+\lceil k/2 \rceil-1}{\lceil k/2 \rceil-1} = \binom{z}{w}$. Lucas's theorem modulo 2 says $\binom{z}{w}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \left\{ \begin{matrix} -k \\ -n \end{matrix} \right\}, \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left[\begin{matrix} -k \\ -n \end{matrix} \right]. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\left[\begin{matrix} -n \\ k \end{matrix} \right] = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\left[\begin{matrix} -5 \\ k \end{matrix} \right] = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x)\cdots(n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$B_{-1} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$. $\square$

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = [t^k] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $O(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$T_{n+1}(x) = x(T_n(x) + T'_n(x)), \quad T_0(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of T_n are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $T_1(x) = x$. Suppose $f = T_n$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + O(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + O(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

Let $v = n/k > 1$ and let $G \in (0, 1)$ be the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Equivalently, if $r = v - G$, then $r/(1 - e^{-r}) = v$. Saddle-point extraction from $(e^z - 1)^k/k!$ gives

$$\left\{ \frac{n}{k} \right\} \sim \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k}, \quad (7.13)$$

uniformly away from the two extreme edges; refined versions have relative error $O(1/n)$, and the numerical constant $0.066/n$ quoted in the archive belongs to one particular explicit error theorem.

Derivation. By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi i k!} \int \exp(k \log(e^z - 1) - n \log z) \frac{dz}{z}.$$

The positive saddle r solves $ke^r/(e^r - 1) = n/r$, or $r/(1 - e^{-r}) = v$. Put $G = v - r$ to obtain Equation (7.12). Expanding the phase to second order at r , evaluating the Gaussian integral, and simplifying with Stirling's formula for $n!/k!$ gives exactly Equation (7.13). On a fixed angular complement of the saddle the real part of the phase drops quadratically; Taylor's theorem with an explicit third- and fourth-derivative bound gives a relative $O(1/n)$ remainder. Thus the formula is a theorem of the saddle method, not a heuristic replacement of the generating function. $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.4 (Stirling-zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n v^{\bar{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i / (e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/v^{\bar{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{v^{\bar{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m} / (e^u - 1) du = \Gamma(z+1-m) \zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\zeta(s, v) = \frac{k!}{s - k} \sum_{n \geq 0} \frac{\begin{bmatrix} n+k \\ n \end{bmatrix}}{(n+k)!} \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.5 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $O(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^w$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z+w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.6 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.22)$$

Proof. Put $L(u) = -\log(1-u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell+1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right], \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \left[\begin{matrix} n \\ r \end{matrix} \right] u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1-u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.5 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell+1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.7 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m = & \frac{1}{2}\delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ & \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^{\infty} \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^{\infty} \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1+w)}{1+w} = m! \sum_{r \geq m} s(r+1, m+1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k+2, m+1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k+1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k+1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k+1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.5 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno’s formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n are defined by

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} G_n z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1+z)$ gives

$$n!G_n = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell+1}. \quad (7.30)$$

Indeed, $z/\log(1+z) = \int_0^1 (1+z)^t dt$ and comparison with $(1+z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n \mu^k s(n, n-k+1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}D_t^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand D_t^n by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N} \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := [x^{k-1}](x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.2 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n + 1$. If deleting $n + 1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n + 1$ has exactly r elements: choose its $r - 1$ companions and partition the remaining $n - r + 1$ labels into $k - 1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j / j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.3 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n - d + 1 \\ k - d + 1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i - j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) q^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d - 1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d - 1$ vertices, which form a clique, and hence has $q - d + 1$ choices. Thus

$$\chi_{G_{n,d}}(q) = q^{d-1} (q - d + 1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned} \chi_{G_{n,d}}(q) &= q^{d-1} \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q - d + 1^r \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r+d-1}. \end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). $\square$

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0 = B_1 = 1, \quad (B_n)_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.4. *If $N = p_1 \cdots p_n$ is squarefree, then B_n is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.5. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.6. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.7 (Return-to-order shuffles). *Exactly B_n of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.8. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.9. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{ccccccc} 1 & & & & & & \\ 1 & 2 & & & & & \\ 2 & 3 & 5 & & & & \\ 5 & 7 & 10 & 15 & & & \\ 15 & 20 & 27 & 37 & 52 & & \end{array}$$

Theorem 9.10. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n$ and the right edge is $a_{n,n} = B_{n+1}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). For $n \geq 0$,

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

10.2 Spivey's identity and binomial inversions

Theorem 10.3 (Spivey's identity). For $m, n \geq 0$,

$$B_{m+n} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.4 (Bell inversion identities). For $n, k \geq 0$,

$$B_n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.7 below gives $\mathbb{E}X^r = B_r$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.5 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}. \quad (10.9)$$

Proof. By Dobiński's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} m^i = m^{\underline{k}}$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{m^{\underline{k}} [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

10.3 Exponential generating function and differential equation

Theorem 10.6. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

10.4 Dobiński's formula and Poisson moments

Theorem 10.7 (Dobiński). *For every $n \geq 0$,*

$$B_n = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z-1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a Poisson(1) variable. $\square$

Corollary 10.8 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. $\square$

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n to the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n(1) = B_n.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}(x) \equiv T_{n+1}(x) + x^p T_n(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p} \equiv B_n + B_{n+1} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Theorem 11.2 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m} \equiv m B_n + B_{n+1} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}(x) \equiv T_{n+1}(x) + T_n(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1} + T_n \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

11.2 A universal period bound modulo a prime

Theorem 11.3. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.4. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.5. *For $n \geq 1$,*

$$1 \leq \frac{B_{n-1}B_{n+1}}{B_n^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n) is log-convex, whereas $(B_n/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1}\mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{e^t - 1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1} z_{j+1}) = j/(j-1)$. Hence $(B_n/n!)^2 \geq (B_{n-1}/(n-1)!)(B_{n+1}/(n+1)!)$, which is the right inequality in (11.6). $\square$

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.6. *For $t > 0$,*

$$B_n \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$B_n^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.7 (Uniform numerical bounds). *For every $n \geq 1$,*

$$B_n < \left(\frac{0.792 n}{\log(n+1)} \right)^n. \quad (11.9)$$

Define, for $x > 1$,

$$d(x) = \log \log(x+1) - \log \log x + \frac{1 + e^{-1}}{\log x}. \quad (11.10)$$

For every $\varepsilon > 0$ there is $n_0(\varepsilon) = \max\{e^4, d^{-1}(\varepsilon)\}$ such that, for $n > n_0(\varepsilon)$,

$$B_n < \left(\frac{e^{-0.6+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.11)$$

Proof. Set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

Direct differentiation, after multiplying by the positive denominator $2x \log x \log(x+1) \log(x/\log(x+1))^2$, shows that $R'(x) < 0$ for $x \geq 39$; substitution gives $R(39) < 0.792$. Thus (11.8) proves (11.9) for $n \geq 39$. The remaining $1 \leq n \leq 38$ are obtained successively from (10.1); substituting them gives strict inequalities. This is a finite exact integer check.

A second differentiation gives, for $x \geq e^4$,

$$\log R(x) \leq -0.6 + d(x), \quad d'(x) < 0, \quad \lim_{x \rightarrow \infty} d(x) = 0.$$

(The numerator after clearing positive denominators is a sum of $-(\log x - 4)^2$, $-(x - 1 - \log x)$, and negative multiples of these two standard nonnegative quantities.) Hence $d(n) < \varepsilon$ when $n > d^{-1}(\varepsilon)$, and (11.11) follows from (11.8). $\square$

11.5 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1+r)$.

Theorem 11.8 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \quad (11.12)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1\right). \quad (11.13)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n}{n!} = \frac{e^{e^r} - 1}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du/\sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.12). Stirling's formula and $e^r = n/r$ simplify it to (11.13). $\square$

The archived refined expansion used undefined symbols. The following construction makes every coefficient unambiguous. Let $T_j(r) = \sum_{q=0}^j \{j\}_q r^q$ be the Touchard polynomial and let $Z \sim N(0, 1)$. For an integer $h = O(r)$ define

$$\mathcal{R}_{r,h}(Z) = \exp\left(-\frac{ihZ}{\sqrt{A}} + \sum_{j \geq 3} \frac{i^j e^r T_j(r)}{j! A^{j/2}} Z^j\right). \quad (11.14)$$

Expand this formal exponential and take Gaussian moments termwise; retaining all monomials of total order at most e^{-Mr} defines $C_0(r, h), \dots, C_M(r, h)$, with $C_0 = 1$.

Theorem 11.9 (Uniform Moser–Wyman scheme). *For every fixed M and $|h| \leq Cr$,*

$$B_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi A}} \left(\sum_{q=0}^{M-1} C_q(r, h) + O_C(e^{-Mr} P_M(r)) \right), \quad (11.15)$$

where P_M is an explicitly computable polynomial. The first correction is

$$C_1(r, h) = -\frac{h^2}{2A} - \frac{he^r T_3(r)}{2A^2} + \frac{e^r T_4(r)}{8A^2} - \frac{5e^{2r} T_3(r)^2}{24A^3}. \quad (11.16)$$

In particular the coefficient of e^{-qr} is a polynomial in h of degree at most $2q$, with rational functions of r as coefficients.

Proof. Use the same circle as in the preceding proof but replace n by $n+h$. After $\theta = Z/\sqrt{A}$, the ratio of the integrand to its Gaussian quadratic part is exactly the formal series (11.14). Watson's lemma with a Gaussian majorant permits termwise integration on the central arc; the complementary arcs have the same exponential bound uniformly for $|h| \leq Cr$. Odd Gaussian moments vanish, so only integral powers of e^{-r} remain. Since $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, the terms from the square of the h -term, its product with the cubic term, the quartic term, and the square of the cubic term give (11.16). The same finite bookkeeping defines every subsequent C_q and bounds the omitted Taylor remainder. $\square$

Corollary 11.10 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n}{n} &= \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ &\quad + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.17)$$

Proof. Take logarithms in (11.13). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + \frac{\log \log n}{\log n} + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $O(\log n/n)$. $\square$

11.6 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0 = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\hat{\mathcal{B}}_{n,k}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\hat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $\hat{X}(t) = \sum_{j \geq 1} x_j t^j$,

$$\hat{X}(t)^k = \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{u\hat{X}(t)} = \sum_{n \geq k \geq 0} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1} &= x_1, \\
\mathcal{B}_{2,1} &= x_2, & \mathcal{B}_{2,2} &= x_1^2, \\
\mathcal{B}_{3,1} &= x_3, & \mathcal{B}_{3,2} &= 3x_1x_2, & \mathcal{B}_{3,3} &= x_1^3, \\
\mathcal{B}_{4,1} &= x_4, & \mathcal{B}_{4,2} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3} &= 6x_1^2x_2, & \mathcal{B}_{4,4} &= x_1^4, \\
\mathcal{B}_{5,1} &= x_5, & \mathcal{B}_{5,2} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4} &= 10x_1^3x_2, & \mathcal{B}_{5,5} &= x_1^5, \\
\mathcal{B}_{6,1} &= x_6, & \mathcal{B}_{6,2} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5} &= 15x_1^4x_2, & \mathcal{B}_{6,6} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0 &= 1, \\
\mathcal{B}_1 &= x_1, \\
\mathcal{B}_2 &= x_1^2 + x_2, \\
\mathcal{B}_3 &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4 &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5 &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6 &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7 &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}, \quad (13.2)$$

with $\mathcal{B}_{0,0} = 1$, $\mathcal{B}_{n,0} = 0$ for $n > 0$, and $\mathcal{B}_{0,k} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}(x) \mathcal{B}_{n-i,k_2}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B (|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1} = x_n, \quad \mathcal{B}_{n,2} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n} = x_1^n, \quad \mathcal{B}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (1.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.5. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t))^2 F(t).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}(1, 1, \dots, 1) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (14.3)$$

$$\mathcal{B}_n(1, 1, \dots, 1) = \mathcal{B}_n, \quad (14.4)$$

$$\mathcal{B}_{n,k}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = L(n, k), \quad (14.5)$$

$$T_n(x) = \mathcal{B}_n(x, x, \dots, x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $T_n(x)$. $\square$

14.2 Homogeneity and a useful evaluation

Theorem 14.2 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}(1, 2, 3, \dots, n-k+1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k}k^{n-k}$. $\square$

14.3 Addition and shifted-zero identities

Theorem 14.3 (Complete Bell addition formula).

$$\mathcal{B}_n(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}(x) \mathcal{B}_i(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.4 (Insertion of q leading zeros). For $q \geq 0$,

$$\begin{aligned} \mathcal{B}_{n,k} \left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots \right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

14.4 Exponential–logarithmic inversion

Theorem 14.5 (Bell transform and its inverse). Suppose

$$y_n = \mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n/n!$. $\square$

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno's formula in Equation (19.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n = (h_{ij})$ and $K_n = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1} x_2 & \binom{n-1}{2} x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1} x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.6 (Bell determinants).

$$\mathcal{B}_n(x_1, \dots, x_n) = \det H_n = \det K_n. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n = D(JH_n^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.7. *If $x^{k\Diamond}$ denotes the k -fold diamond power, then*

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k\Diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k\Diamond}$ is $X(t)^k$. Compare with (1.10). □

For example,

$$x\Diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3\Diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n \sum_{k=1}^n \frac{1}{k} \widehat{\mathcal{B}}_{n,k}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k = -(k-1)! \operatorname{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n(s_1, s_2, \dots, s_n). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \operatorname{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.5. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n , $\mathcal{B}_n$, $\beta_n(x)$, and C_n are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$A_0(t) = 1, \quad A_n(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$A_n(t) = (1 + (n-1)t)A_{n-1}(t) + t(1-t)A'_{n-1}(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1-\pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad A_n(t) = t^{n-1}A_n(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} A_n(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.4. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Theorem 16.5 (A second polynomial recurrence). For $n > 1$,

$$A_n(t) = \sum_{k=0}^{n-1} \binom{n}{k} A_k(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

16.3 Worpitzky's identity and sums of powers

Theorem 16.6 (Worpitzky). For every indeterminate x and $n \geq 1$,

$$x^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Corollary 16.7 (Power sums). For $m, n \geq 1$,

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z A_n(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.8.

$$A_n(t) = \sum_{k=0}^n \left\{ n \atop k \right\} k! (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\{ n \atop k \right\} m^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} m^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $A_n(t) = \sum_k \left\{ n \atop k \right\} k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.9. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0, 1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle n \atop k \right\rangle / n!$.

Proof. The volume of $\{x \in [0, 1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle n \atop k \right\rangle$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.10. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n}{k}} = (n+1) \beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n A_n \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $A_n(t)$, so its h^* -vector is the n th Eulerian row.*

Proof. By the defining Ehrhart identity and the lattice-point count above,

$$\frac{h_C^*(t)}{(1-t)^{n+1}} = \sum_{m \geq 0} (m+1)^n t^m.$$

Equation (16.6) identifies the right side with $A_n(t)/(1-t)^{n+1}$. Multiplication by $(1-t)^{n+1}$ therefore gives $h_C^*(t) = A_n(t)$ coefficient by coefficient. $\square$

The regions $\Delta_{n,k}$ in Theorem 16.9 are half-open hypersimplices. Their normalized volumes are $n! \operatorname{vol} \Delta_{n,k} = \langle \frac{n}{k} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \frac{n}{k} \right\}$ ordered partitions into k blocks.

Theorem 17.2. *The h -polynomial of the simple polytope dual to the type- A permutohedron is $A_n(t)$.*

Proof. For a simple $(n-1)$ -polytope the h -polynomial is the transform $h(t) = \sum_F (t-1)^{\dim F}$ over faces of the dual simplicial complex. Grouping braid-fan faces by their k ordered blocks gives $\sum_k k! \left\{ \frac{n}{k} \right\} (t-1)^{n-k}$, which is $A_n(t)$ by (16.12). $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_B(n, k)$ count signed permutations with k such descents, and set $M_n(t) = \sum_{k=0}^n E_B(n, k)t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_B(n, k) = (2k + 1)E_B(n - 1, k) + (2n - 2k + 1)E_B(n - 1, k - 1), \quad (18.1)$$

$$\sum_{m \geq 0} (2m + 1)^n t^m = \frac{M_n(t)}{(1 - t)^{n+1}}, \quad (18.2)$$

$$E_B(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j) + 1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n - 1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k + 1$ preserving slots for an object with k descents and $2n - 2k + 1$ creating slots for one with $k - 1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m + 1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m + 1)^n = \sum_{k=0}^n E_B(n, k) \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1 - t)^{n+1}$ to obtain (18.2). Multiplication by $(1 - t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

As in type A , these coefficients form the h -vector of the simple polytope dual to the type- B permutohedron; the proof repeats the braid-fan argument with signed ordered partitions.

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$P_n(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad P_0(t) = 1.$$

Theorem 18.2 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n-1 \rangle\rangle_{k-1} + (k+1) \langle\langle n-1 \rangle\rangle_k, \quad (18.4)$$

$$P_{n+1}(t) = (2nt + 1)P_n(t) - t(t-1)P'_n(t), \quad (18.5)$$

$$P_n(1) = (2n-1)!! \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n-1$ gaps of a Stirling permutation of order $n-1$. A descent gap or the terminal gap preserves the number of descents, giving $k+1$ choices from an object with k descents. Each of the remaining $2n-k-1$ gaps creates one descent from an object with $k-1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n-1$ insertion gaps, so the total count is $(2n-1)!!$. $\square$

The differential recurrence has two useful equivalent forms:

$$(t-1)^{-2n-2}P_{n+1}(t) = (t(1-t)^{-2n-1}P_n(t))', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1-t} u_n(t) \right)', \quad u_n(t) = (t-1)^{-2n}P_n(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.3 (Diagonal Stirling generating function). For $n \geq 1$,

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n+m \\ m \end{matrix} \right\} t^m = \frac{tP_n(t)}{(1-t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1-t)F_{n+1} = tF'_n$, with $F_0 = (1-t)^{-1}$. Hence $F_1 = t/(1-t)^3$. If the displayed formula holds for n , differentiating $tP_n/(1-t)^{2n+1}$ and using (18.5) gives the formula for $n+1$. $\square$

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.4 (Second-order Eulerian diagonal formula). For $p, m \geq 0$,

$$\left[\begin{matrix} m \\ m-p \end{matrix} \right] = \sum_{j=0}^p \langle\langle p \rangle\rangle_j \binom{m+j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\begin{bmatrix} x \\ x-p \end{bmatrix} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \begin{bmatrix} x \\ x-1 \end{bmatrix} &= \binom{x}{2}, \\ \begin{bmatrix} x \\ x-2 \end{bmatrix} &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \begin{bmatrix} x \\ x-3 \end{bmatrix} &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \begin{bmatrix} m \\ m-p \end{bmatrix} t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} P_p(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2 ((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} P_p(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $P_p(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Differentiation, Composition, and Reversion

Chapter 19

Faà di Bruno's formula

19.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 19.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (19.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

19.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 19.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (19.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (19.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (19.1) gives (19.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (19.3). $\square$

For $n = 4$ this reads

$$\begin{aligned} (f \circ g)^{(4)} &= f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 \\ &\quad + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \end{aligned} \quad (19.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

19.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (19.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (19.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

19.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial/\partial x_j$.

Theorem 19.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (19.6)$$

Proof. The proof of Theorem 19.1 applies verbatim, except that the new derivative is $\partial/\partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (19.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

Chapter 20

Composition of ordinary and exponential series

20.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 20.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \dots b_{i_k}, \quad (20.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \dots b_n^{\pi_n} \quad (20.2)$$

$$= \sum_{k=1}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad (20.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \widehat{\mathcal{B}}_{n,k}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (20.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

20.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 20.2 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}). \quad (20.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (20.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp \left(\sum_{j \geq 1} a_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (20.7)$$

If the series converge, (20.5) is exactly (19.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 21

Lagrange inversion and Lagrange–Bürmann

21.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (21.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 21.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n]H(g(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n, \quad (21.2)$$

$$= [w^n]H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (21.3)$$

In particular,

$$[z^n]g(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n. \quad (21.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.2 give

$$\begin{aligned} [z^n]H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1}f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (21.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (21.3). Take $H(w) = w$ for (21.4). □

21.2 Analytic inverse functions

Theorem 21.2 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (21.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (21.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta = z - b$, and

$$\phi(u) = \frac{u}{f(a + u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta \phi(u)$. Formula (21.4), multiplied by $n!$, is exactly (21.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + O((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

21.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$\hat{f}_j = \frac{f_{j+1}}{(j+1)f_1}. \quad (21.7)$$

Theorem 21.3 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k n^{\bar{k}} \mathcal{B}_{n-1,k}(\hat{f}_1, \dots, \hat{f}_{n-k}). \quad (21.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} \hat{f}_j \frac{w^j}{j!}.$$

By (21.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1 + U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

21.4 The associahedron interpretation, with normalized notation

The formula in the archive mixes exponential coefficients with an ordinary-series face formula. The exact geometric statement is the following.

Let

$$A(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad B(x) = A^{\langle -1 \rangle}(x) = x + \sum_{n \geq 1} b_n x^{n+1}. \quad (21.9)$$

Use the indexing in which the associahedron K_n has dimension $n - 1$ and its faces are plane rooted trees with $n + 1$ leaves and internal arities at least 2. If an internal vertex has arity $i + 1$, give it weight a_i ; let a_F be the product of all vertex weights. Equivalently, if $F \cong K_{i_1} \times \cdots \times K_{i_m}$, then $a_F = a_{i_1} \cdots a_{i_m}$.

Theorem 21.4 (Associahedral inversion).

$$b_n = \sum_{F \text{ a face of } K_n} (-1)^{n - \dim F} a_F. \quad (21.10)$$

Proof. The equation $A(B) = x$ is

$$B = x - \sum_{i \geq 1} a_i B^{i+1}.$$

Iterating it recursively expands b_n as a sum over plane rooted trees with $n + 1$ leaves: every internal vertex of arity $i + 1$ contributes $-a_i$. A face F of K_n is exactly such a tree after contracting a chosen set of internal edges. If it has m internal vertices, then $\dim F = n - m$, so its sign is $(-1)^m = (-1)^{n - \dim F}$ and its weight is a_F . Thus the tree expansion is the alternating face sum. $\square$

For example,

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_2a_1 - 5a_1^3, \\ b_4 &= -a_4 + 6a_3a_1 + 3a_2^2 - 21a_2a_1^2 + 14a_1^4. \end{aligned}$$

For K_4 , the coefficients 1, 6, 3, 21, 14 count respectively the whole 3-polytope, its six pentagonal and three square facets, its edges, and its vertices.

Chapter 22

Algebraic and transcendental examples

22.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (22.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 22.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (22.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (22.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (21.4),

$$[z^n]x(z) = \frac{1}{n}[w^{n-1}](1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

22.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 22.2. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + O(z^5), \quad (22.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (21.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

22.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (22.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + O(\xi^6). \quad (22.6)$$

Proof. The derivative at $u = 0$ of the left side of (22.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2$, $c_2 = 1/16$, $c_3 = -1/192$, $c_4 = -1/3072$, $c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143\dots$

22.4 Binary trees and Catalan numbers

Let C_n count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (22.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n = [z^n]C(z) = \frac{1}{n}[w^{n-1}](1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (22.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n is used here to avoid confusing these binary-tree numbers with Bell numbers.

Chapter 23

Lagrange's reversion theorem in shifted form

23.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (23.1)$$

Theorem 23.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (23.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (23.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (21.2) to $H(u) = g(x + u)$:

$$[y^k]g(v) = \frac{1}{k} [u^{k-1}]g'(x + u)f(x + u)^k = \frac{1}{k!} D_x^{k-1}(g'(x)f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

23.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x)g(z)(1 - yf'(z)) \, dz. \quad (23.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (23.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (23.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (23.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 24

Laplace-type asymptotic integrals

24.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (24.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (24.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (24.3)$$

Let $\widehat{\mathcal{B}}_{k,j}$ denote the ordinary Bell polynomial.

Theorem 24.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (24.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (24.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = tA(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s))t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} [t^k] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (21.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k]A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (24.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 24.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VII

Incidence Algebra, Umbral Calculus, and Multivariate Inversion

Chapter 25

Binomial inversion, partition lattices, and Sheffer–Riordan calculus

The concrete triangular arrays developed earlier are instances of a larger calculus. This chapter collects the genuinely nonduplicate material from the five drafts: incidence-algebra inversion, the partition-lattice Möbius function, Sheffer sequences, and exponential Riordan arrays. These viewpoints explain why changes of basis, labelled decomposition, and formal substitution repeatedly lead to the same matrices.

25.1 Binomial inversion

Theorem 25.1 (Binomial inversion). *For sequences (a_n) and (b_n) in a commutative $\mathbb{Q}$ -algebra, the formulas*

$$b_n = \sum_{k=0}^n \binom{n}{k} a_k, \quad (25.1)$$

$$a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k \quad (25.2)$$

are equivalent. In exponential generating functions,

$$B(z) = e^z A(z), \quad A(z) = e^{-z} B(z). \quad (25.3)$$

Proof. Multiplication of EGFs gives

$$e^z A(z) = \sum_{n \geq 0} \left(\sum_{k=0}^n \binom{n}{k} a_k \right) \frac{z^n}{n!},$$

so (25.1) is equivalent to the first identity in (25.3). Multiplication by e^{-z} and coefficient extraction gives (25.2). Directly, substitution of the forward formula into the backward one makes the coefficient of a_j

$$\sum_{k=j}^n (-1)^{n-k} \binom{n}{k} \binom{k}{j} = \binom{n}{j} \sum_{r=0}^{n-j} (-1)^{n-j-r} \binom{n-j}{r} = \delta_{nj},$$

which independently verifies the inverse matrices. □

25.2 The partition lattice and cumulants

Let Π_n be the lattice of partitions of $[n]$, ordered by refinement, with bottom $\widehat{0}$ and top $\widehat{1}$.

Theorem 25.2 (Möbius function of Π_n). *If $\sigma \leq \pi$ and m_C is the number of blocks of σ contained in a block C of π , then*

$$\mu(\sigma, \pi) = \prod_{C \in \pi} (-1)^{m_C-1} (m_C - 1)!. \quad (25.4)$$

In particular,

$$\mu(\widehat{0}, \pi) = \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)!, \quad (25.5)$$

$$\mu(\pi, \widehat{1}) = (-1)^{|\pi|-1} (|\pi| - 1)!. \quad (25.6)$$

Proof. The interval $[\sigma, \pi]$ is the direct product, over $C \in \pi$, of partition lattices on the m_C blocks of σ inside C . Möbius functions multiply on direct products, so it remains to compute $u_m = \mu_{\Pi_m}(\widehat{0}, \widehat{1})$. The defining relation gives, for $m > 1$,

$$0 = \sum_{\rho \in \Pi_m} \mu(\widehat{0}, \rho) = \sum_{\rho \in \Pi_m} \prod_{B \in \rho} u_{|B|}.$$

Let $U(z) = \sum_{m \geq 1} u_m z^m / m!$. By the labelled exponential formula,

$$\exp U(z) = 1 + z,$$

because the inner sum is 1 for $m = 0, 1$ and 0 thereafter. Hence $U(z) = \log(1 + z)$ and $u_m = (-1)^{m-1} (m - 1)!$. Product decomposition now gives the general formula. $\square$

Theorem 25.3 (Moment–cumulant inversion). *The relations*

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad (25.7)$$

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|} \quad (25.8)$$

are inverse. Equivalently, if $M(z) = 1 + \sum_{n \geq 1} m_n z^n / n!$ and $K(z) = \sum_{n \geq 1} \kappa_n z^n / n!$, then $M = e^K$ and $K = \log M$.

Proof. The forward relation is the exponential formula for a set of blocks weighted by cumulants. Möbius inversion on Π_n , using (25.6), gives the backward relation. Grouping partitions by block sizes is exactly the coefficient expansion of e^K and $\log M$. $\square$

25.3 Sheffer sequences

Definition 25.4. A polynomial sequence $(s_n(x))_{n \geq 0}$ is Sheffer for (A, B) when

$$\sum_{n \geq 0} s_n(x) \frac{t^n}{n!} = A(t) e^{xB(t)}, \quad (25.9)$$

where $A(0) \neq 0$, $B(0) = 0$, and $B'(0) \neq 0$. It is of binomial type when $A = 1$, and Appell when $B(t) = t$.

Theorem 25.5 (Addition and lowering). *Let p_n be the binomial-type sequence with EGF $e^{yB(t)}$. Then*

$$s_n(x+y) = \sum_{k=0}^n \binom{n}{k} s_k(x) p_{n-k}(y). \quad (25.10)$$

If $\overline{B}$ is the compositional inverse of B and $Q = \overline{B}(D_x)$, then

$$Qs_n(x) = ns_{n-1}(x). \quad (25.11)$$

Proof. Multiplying the two EGFs at x and y proves the addition law. For every formal series F , $F(D_x)e^{xB(t)} = F(B(t))e^{xB(t)}$. Taking $F = \overline{B}$ gives the eigenvalue t ; multiplication of the EGF by t and coefficient comparison gives the lowering law. $\square$

Proposition 25.6 (Abel polynomials). *Let $T(t)$ solve $T = te^{-aT}$. Then*

$$e^{xT(t)} = 1 + \sum_{n \geq 1} x(x-na)^{n-1} \frac{t^n}{n!}. \quad (25.12)$$

Thus $x(x-na)^{n-1}$ is a sequence of binomial type.

Proof. Lagrange–Bürmann gives

$$[t^n]e^{xT(t)} = \frac{x}{n} [w^{n-1}]e^{xw}e^{-anw} = \frac{x(x-na)^{n-1}}{n!}.$$

An EGF of the form $e^{xB(t)}$ satisfies the binomial identity by multiplication at x and y . $\square$

25.4 Exponential Riordan arrays

Definition 25.7. For $g(0) \neq 0$, $f(0) = 0$, and $f'(0) \neq 0$, define

$$R_{n,k} = \frac{n!}{k!} [t^n]g(t)f(t)^k. \quad (25.13)$$

The resulting lower-triangular matrix is denoted $[g, f]$.

Theorem 25.8 (Riordan action, product, and inverse). *If (a_k) has EGF $A(t)$, then $[g, f]a$ has EGF*

$$g(t)A(f(t)). \quad (25.14)$$

Consequently,

$$[g, f][h, \ell] = [g(h \circ f), \ell \circ f], \quad (25.15)$$

$$[g, f]^{-1} = \left[\frac{1}{g \circ \overline{f}}, \overline{f} \right]. \quad (25.16)$$

Proof. By definition,

$$\sum_n \left(\sum_k R_{n,k} a_k \right) \frac{t^n}{n!} = g(t) \sum_k a_k \frac{f(t)^k}{k!} = g(t)A(f(t)).$$

Applying this action twice gives the product law, and substituting $\overline{f}$ gives the inverse. $\square$

Example 25.9 (Stirling matrices). The second-kind matrix is $[1, e^t - 1]$, while the signed first-kind matrix is $[1, \log(1+t)]$. Since the two substitution series are compositional inverses, Stirling inversion is the Riordan inverse law. The Bell-polynomial matrix is obtained by replacing $e^t - 1$ with $\sum_{j \geq 1} x_j t^j / j!$.

Chapter 26

Complementary Bell numbers and signed partition calculus

The MathWorld material includes the complementary Bell numbers, which are the set-partition transform at block weight -1 . They are especially useful as a clean example of signed labelled enumeration.

Definition 26.1. Define

$$\widetilde{B}_n = \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (26.1)$$

Thus $\widetilde{B}_0 = 1$ and the next values are $-1, 0, 1, 1, -2, -9, -9, 50, 267, 413, \dots$

Theorem 26.2 (EGF, recurrence, and Dobinski-type series). *The complementary Bell numbers satisfy*

$$\sum_{n \geq 0} \widetilde{B}_n \frac{z^n}{n!} = \exp(1 - e^z), \quad (26.2)$$

$$\widetilde{B}_{n+1} = - \sum_{k=0}^n \binom{n}{k} \widetilde{B}_k, \quad (26.3)$$

$$\widetilde{B}_n = e \sum_{m=0}^{\infty} \frac{(-1)^m m^n}{m!}. \quad (26.4)$$

The last series is absolutely convergent.

Proof. Using the EGF of the Stirling numbers and summing over k gives

$$\sum_{n \geq 0} \widetilde{B}_n \frac{z^n}{n!} = \sum_{k \geq 0} \frac{(-1)^k}{k!} (e^z - 1)^k = e^{1-e^z}.$$

Differentiating yields $C'(z) = -e^z C(z)$; the exponential Cauchy product gives the recurrence. Finally expand e^{-e^z} :

$$e^{1-e^z} = e \sum_{m \geq 0} \frac{(-1)^m}{m!} e^{mz} = \sum_{n \geq 0} \left(e \sum_{m \geq 0} \frac{(-1)^m m^n}{m!} \right) \frac{z^n}{n!}.$$

Absolute convergence follows from the ratio test for $m^n/m!$ at fixed n . □

Corollary 26.3 (Convolution inverse of the Bell sequence). *For every $n \geq 0$,*

$$\sum_{k=0}^n \binom{n}{k} B_k \widetilde{B}_{n-k} = \delta_{n0}. \quad (26.5)$$

Proof. The two EGFs are e^{e^z-1} and e^{1-e^z} , whose product is 1. □

Remark 26.4 (Signed combinatorial meaning). Equation (26.1) is the signed sum of all partitions of $[n]$, with weight $(-1)^{\# \text{blocks}}$. It is not a count of an ordinary unweighted class; it is an Euler characteristic in the incidence-algebra sense. The convolution inverse is the exponential formula applied to component weights $+1$ and -1 .

Chapter 27

Coordinate-free composition and derivatives of inverse maps

27.1 Fréchet Faà di Bruno formula

Theorem 27.1 (Coordinate-free Faà di Bruno). *Let E, F, G be Banach spaces, $g : E \rightarrow F$, $f : F \rightarrow G$, and let D^r denote the symmetric r -linear Fréchet derivative. Then*

$$D^n(f \circ g)(x)[h_1, \dots, h_n] = \sum_{\pi \in \Pi_n} D^{|\pi|} f(g(x)) \left[D^{|B|} g(x)[h_i : i \in B] \right]_{B \in \pi}. \quad (27.1)$$

Proof. Induct on n . For $n = 1$ this is the chain rule. Differentiate a term indexed by $\pi \in \Pi_n$ in direction h_{n+1} . Either the derivative hits the outer derivative, creating the new singleton block $\{n+1\}$, or it hits one inner derivative associated with a block B , adjoining $n+1$ to B . Every partition of $[n+1]$ is obtained uniquely in one of these two ways, according as $\{n+1\}$ is a singleton or not. The product and chain rules for multilinear maps supply no further multiplicities because blocks are already labelled subsets. $\square$

Corollary 27.2. *When all directions are equal to h ,*

$$D^n(f \circ g)(x)[h^n] = \sum_{k=1}^n D^k f(g(x)) [\mathcal{B}_{n,k}(Dg[h], D^2g[h^2], \dots)], \quad (27.2)$$

where the Bell polynomial is interpreted as the symmetric tensor sum over blocks. In one dimension this is the usual scalar Faà di Bruno formula.

Proof. Group the set partitions in (27.1) by their number of blocks and then by block-size multiplicities. The number of partitions of a fixed profile is exactly the coefficient defining $\mathcal{B}_{n,k}$. $\square$

27.2 Recursive derivatives of an inverse function

Theorem 27.3 (Inverse derivative recursion). *Let $g = f^{-1}$ locally and $f'(g(y)) \neq 0$. For $n \geq 2$,*

$$g^{(n)}(y) = -\frac{1}{f'(g(y))} \sum_{k=2}^n f^{(k)}(g(y)) \mathcal{B}_{n,k}(g'(y), g''(y), \dots, g^{(n-k+1)}(y)). \quad (27.3)$$

In particular, writing all derivatives of f at $x = g(y)$,

$$g' = \frac{1}{f'}, \quad (27.4)$$

$$g'' = -\frac{f''}{(f')^3}, \quad (27.5)$$

$$g''' = \frac{3(f'')^2 - f' f'''}{(f')^5}, \quad (27.6)$$

$$g^{(4)} = \frac{-15(f'')^3 + 10f' f'' f''' - (f')^2 f^{(4)}}{(f')^7}. \quad (27.7)$$

Proof. Differentiate $f(g(y)) = y$ n times. Faà di Bruno gives zero for $n \geq 2$. The term with one outer derivative is $f'(g)\mathcal{B}_{n,1} = f'(g)g^{(n)}$; every other term involves only derivatives of g of order at most $n - 1$. Isolating the first term proves the recursion. The displayed cases follow by substituting $\mathcal{B}_{2,2} = x_1^2$, $\mathcal{B}_{3,2} = 3x_1x_2$, $\mathcal{B}_{4,2} = 4x_1x_3 + 3x_2^2$, and $\mathcal{B}_{4,3} = 6x_1^2x_2$, then simplifying. $\square$

27.3 Several nested functions

For a chain $f_1 \circ \cdots \circ f_r$, repeated use of Theorem 27.1 produces a levelled hierarchy of set partitions. At the innermost level, derivative labels that strike the same occurrence of f_r form blocks; at the next level, those blocks are grouped according to occurrences of f_{r-1} , and so on. Flattening the hierarchy in either association order is canonical, giving a combinatorial proof of associativity of formal composition. For three scalar functions,

$$D^n f(g(h(x))) = \sum_{k=0}^n f^{(k)}(g(h(x))) \mathcal{B}_{n,k}((g \circ h)', \dots, (g \circ h)^{(n-k+1)}), \quad (27.8)$$

$$(g \circ h)^{(j)} = \sum_{\ell=0}^j g^{(\ell)}(h(x)) \mathcal{B}_{j,\ell}(h', \dots, h^{(j-\ell+1)}). \quad (27.9)$$

Substitution of the second formula into the first is a finite closed formula; the partition hierarchy explains its multiplicities without additional index machinery.

Chapter 28

Good's multivariate Lagrange inversion formula

Let $\mathbf{w} = (w_1, \dots, w_d)$ be determined formally by

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0. \quad (28.1)$$

For a multi-index $\mathbf{n}$, write $\mathbf{t}^{\mathbf{n}} = \prod_i t_i^{n_i}$.

Lemma 28.1 (Multivariate residue substitution). *Let*

$$T_i(\mathbf{x}) = x_i U_i(\mathbf{x}), \quad U_i(\mathbf{0}) \neq 0 \quad (1 \leq i \leq d).$$

Then, for every formal Laurent series for which either side is defined coefficientwise,

$$\operatorname{Res}_{\mathbf{t}} H(\mathbf{t}) dt_1 \cdots dt_d = \operatorname{Res}_{\mathbf{x}} H(\mathbf{T}(\mathbf{x})) \det \left(\frac{\partial T_i}{\partial x_j} \right) dx_1 \cdots dx_d. \quad (28.2)$$

Proof. Interpret the determinant times $dx_1 \cdots dx_d$ as the pullback of the top differential form $dt_1 \wedge \cdots \wedge dt_d$. By linearity it is enough to take $H(\mathbf{t}) = t_1^{m_1} \cdots t_d^{m_d}$. If $m_j \neq -1$ for some j , then

$$\mathbf{t}^{\mathbf{m}} dt_1 \wedge \cdots \wedge dt_d = d \left(\frac{(-1)^{j-1}}{m_j + 1} t_j^{m_j+1} \prod_{i \neq j} t_i^{m_i} dt_1 \wedge \cdots \wedge \widehat{dt_j} \wedge \cdots \wedge dt_d \right).$$

Pullback commutes with the exterior derivative. The residue of an exact top form is zero: after expansion it is a sum of formal partial derivatives, and $\partial/\partial x_j$ cannot create an x_j^{-1} term. Thus both sides of (28.2) vanish for this monomial.

It remains to take $m_1 = \cdots = m_d = -1$. Since every U_i is a unit,

$$\frac{dT_i}{T_i} = \frac{dx_i}{x_i} + \frac{dU_i}{U_i}.$$

In $\bigwedge_{i=1}^d dT_i/T_i$, the product $\bigwedge_i dx_i/x_i$ has residue 1. Every other exterior-product term contains at least one logarithmic differential dU_i/U_i , whose coefficients are ordinary power series, and consequently lacks at least one of the negative powers x_j^{-1} needed for the multivariate residue. Its residue is therefore zero. The transformed residue is 1, equal to the residue of $\prod_i t_i^{-1} dt_i$, and the proof is complete. $\square$

Theorem 28.2 (Lagrange–Good inversion). *For every formal series F ,*

$$\begin{aligned} [\mathbf{t}^{\mathbf{n}}] F(\mathbf{w}(\mathbf{t})) &= [\mathbf{x}^{\mathbf{n}}] F(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \\ &\quad \times \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \end{aligned} \quad (28.3)$$

Proof. Represent the left side as

$$\operatorname{Res}_{\mathbf{t}} F(\mathbf{w}(\mathbf{t})) \prod_i t_i^{-n_i-1} dt_1 \cdots dt_d.$$

Use the inverse substitution $t_i = x_i/\phi_i(\mathbf{x})$. Its Jacobian entries are

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

After multiplying by $\prod_i t_i^{-n_i-1}$, the row factors $1/\phi_i$ cancel one power from each $t_i^{-n_i-1}$, leaving $\prod_i \phi_i^{n_i}$. The remaining determinant initially has x_i in row i . Write it as $\det(I - AB)$ with $A = \operatorname{diag}(x_1, \dots, x_d)$ and $B = (\partial_j \log \phi_i)_{i,j}$; Sylvester's identity $\det(I - AB) = \det(I - BA)$ moves the factor x_i from row i to column j , producing the displayed determinant. The remaining residue is exactly the coefficient on the right. $\square$

Example 28.3 (A coupled Fuss–Catalan system). Let

$$u = x(1 + u + v)^a, \quad v = y(1 + u + v)^b,$$

and write $S = 1 + u + v$. For integers $m, n \geq 0$ and arbitrary c , the constant coefficient is $[x^0 y^0] S^c = 1$, while for $m + n \geq 1$,

$$[x^m y^n] S^c = \frac{c}{m!n!} (am + bn + c - 1)^{\overline{m+n-1}}. \quad (28.4)$$

When $am + bn + c \neq 0$, this is equivalently

$$[x^m y^n] S^c = \frac{c}{am + bn + c} \frac{(am + bn + c)^{\overline{m+n}}}{m!n!}. \quad (28.5)$$

Proof. Apply Theorem 28.2 with $\phi_1 = S^a$, $\phi_2 = S^b$, and $F = S^c$. Put $A = am + bn + c$. The determinant is

$$\det \begin{pmatrix} 1 - au/S & -av/S \\ -bu/S & 1 - bv/S \end{pmatrix} = 1 - \frac{au + bv}{S}.$$

Thus the desired coefficient is

$$[u^m v^n] (S^A - auS^{A-1} - bvS^{A-1}).$$

For $m + n \geq 1$, multinomial extraction (with a term containing u or v understood as zero when its requested exponent is zero) gives, after putting all three terms over the common factor $(A - 1)^{\overline{m+n-1}}/(m!n!)$,

$$\frac{(A - 1)^{\overline{m+n-1}}}{m!n!} (A - am - bn) = \frac{c}{m!n!} (A - 1)^{\overline{m+n-1}}.$$

This proves (28.4), including values where $A = 0$ by polynomial continuation. Multiplying and dividing by A when $A \neq 0$ gives (28.5). The case $m = n = 0$ is immediate from $S(0, 0) = 1$. $\square$

Proposition 28.4 (Characteristic determinant). *For the analytic system $H_i(\mathbf{w}, \mathbf{t}) = w_i - t_i \phi_i(\mathbf{w})$, the Jacobian with respect to $\mathbf{w}$, evaluated on a solution, has determinant equal to the Good determinant. Therefore an analytic branch can fail the implicit-function test only where some ϕ_i is singular or this determinant vanishes.*

Proof. Differentiate: $\partial H_i / \partial w_j = \delta_{ij} - t_i \partial_j \phi_i$. On a solution $t_i = w_i / \phi_i(\mathbf{w})$, giving the row form of the determinant; Sylvester's identity converts it to the column form in (28.3). $\square$

Remark 28.5. The determinant identifies candidate branch points, not automatically the radius of every derived series. One must also establish accessibility from the selected branch, exclude earlier coefficient singularities, and check that the outer function F does not cancel the local singular term. This is the multivariate analogue of examining critical values of a one-variable inverse map; see also Bender and Richmond's asymptotic formulation [2].

Part VIII

Bernoulli–Euler Calculus and Special Functions

Chapter 29

Bernoulli, Euler, Genocchi, and Cauchy–Gregory families

Bernoulli numbers already appeared in the Eulerian and power-sum chapters. The MathWorld material adds the surrounding polynomial calculus, Euler and Genocchi families, Bernoulli polynomials of the second kind, and the Euler–Maclaurin formula. These families also illustrate the Sheffer framework: ordinary Bernoulli and Euler polynomials are Appell sequences, whereas the second-kind Bernoulli polynomials are Sheffer for the forward difference operator. Our convention is $\beta_1 = -1/2$, in agreement with the DLMF [16].

29.1 Generating functions and Appell structure

Definition 29.1. The Bernoulli numbers β_n , Bernoulli polynomials $\beta_n(x)$, and Euler polynomials $E_n(x)$ are defined by

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad (29.1)$$

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} \beta_n(x) \frac{t^n}{n!}, \quad (29.2)$$

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} E_n(x) \frac{t^n}{n!}. \quad (29.3)$$

Theorem 29.2 (Appell, translation, and explicit formulas). For $n \geq 1$,

$$\beta'_n(x) = n\beta_{n-1}(x), \quad E'_n(x) = nE_{n-1}(x), \quad (29.4)$$

$$\beta_n(x + y) = \sum_{k=0}^n \binom{n}{k} \beta_k(x) y^{n-k}, \quad E_n(x + y) = \sum_{k=0}^n \binom{n}{k} E_k(x) y^{n-k}, \quad (29.5)$$

$$\beta_n(x) = \sum_{k=0}^n \binom{n}{k} \beta_k x^{n-k}. \quad (29.6)$$

Proof. Differentiating either polynomial EGF with respect to x multiplies it by t , which proves the derivative identities. Replacing x by $x + y$ multiplies by e^{yt} , and the exponential Cauchy product gives translation. Setting $x = 0$ in the Bernoulli translation formula gives the explicit polynomial expansion. $\square$

Theorem 29.3 (Difference and reflection identities). *For all relevant n ,*

$$\beta_n(x+1) - \beta_n(x) = nx^{n-1} \quad (n \geq 1), \quad (29.7)$$

$$\mathbf{E}_n(x+1) + \mathbf{E}_n(x) = 2x^n, \quad (29.8)$$

$$\beta_n(1-x) = (-1)^n \beta_n(x), \quad (29.9)$$

$$\mathbf{E}_n(1-x) = (-1)^n \mathbf{E}_n(x). \quad (29.10)$$

Consequently $\beta_{2m+1} = 0$ for $m \geq 1$ and $\mathbf{E}_{2m+1}(1/2) = 0$.

Proof. Subtract the Bernoulli EGFs at $x+1$ and x :

$$\frac{te^{(x+1)t} - te^{xt}}{e^t - 1} = te^{xt}.$$

Add the two Euler EGFs to obtain $2e^{xt}$; coefficient comparison proves the difference equations. Replacing t by $-t$ in the Bernoulli EGF gives

$$\frac{(-t)e^{-xt}}{e^{-t} - 1} = \frac{te^{(1-x)t}}{e^t - 1},$$

and the analogous Euler calculation proves both reflections. At $x = 0$, $\beta_n(1) = \beta_n$ for $n > 1$ by the difference equation; reflection then forces every odd β_n beyond β_1 to vanish. Euler reflection at $x = 1/2$ proves the last assertion. $\square$

29.2 Power sums and multiplication formulas

Theorem 29.4 (Faulhaber with an offset). *For $N, p \in \mathbb{N}$,*

$$\sum_{j=0}^{N-1} (x+j)^p = \frac{\beta_{p+1}(x+N) - \beta_{p+1}(x)}{p+1}. \quad (29.11)$$

Proof. Apply (29.7) to $\beta_{p+1}(x+j)$ and sum; the left side telescopes. $\square$

Theorem 29.5 (Alternating power sums). *For $N, p \in \mathbb{N}$,*

$$\sum_{j=0}^{N-1} (-1)^j (x+j)^p = \frac{\mathbf{E}_p(x) + (-1)^{N-1} \mathbf{E}_p(x+N)}{2}. \quad (29.12)$$

The formula gives zero when $N = 0$.

Proof. Multiply the Euler difference equation at $x+j$ by $(-1)^j$ and sum. All interior Euler terms cancel in pairs, leaving the two endpoints. $\square$

Theorem 29.6 (Raabe multiplication). *For every positive integer q ,*

$$\sum_{r=0}^{q-1} \beta_n\left(x + \frac{r}{q}\right) = q^{1-n} \beta_n(qx). \quad (29.13)$$

In particular,

$$\beta_n\left(\frac{1}{2}\right) = (2^{1-n} - 1) \beta_n. \quad (29.14)$$

Proof. Sum the generating functions and use the finite geometric series:

$$\sum_{r=0}^{q-1} \frac{te^{(x+r/q)t}}{e^t - 1} = \frac{te^{xt}}{e^{t/q} - 1} = q \frac{(t/q)e^{(qx)(t/q)}}{e^{t/q} - 1}.$$

Coefficient comparison gives the multiplication formula. Set $q = 2$, $x = 0$ and solve for the half-value. $\square$

29.3 Genocchi numbers

Definition 29.7. The Genocchi numbers G_n are defined by

$$\frac{2t}{e^t + 1} = \sum_{n \geq 1} G_n \frac{t^n}{n!}. \quad (29.15)$$

Theorem 29.8 (Bernoulli–Euler–Genocchi relations). For $n \geq 1$,

$$G_n = 2(1 - 2^n)\beta_n = nE_{n-1}(0). \quad (29.16)$$

Thus $G_1 = 1$, $G_{2m+1} = 0$ for $m \geq 1$, and $G_2 = -1$, $G_4 = 1$, $G_6 = -3$, $G_8 = 17$, $G_{10} = -155$.

Proof. Use

$$\frac{2t}{e^t + 1} = \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1}$$

and compare Bernoulli coefficients. Multiplying the Euler EGF at $x = 0$ by t gives the second equality. Parity and initial values follow immediately. $\square$

29.4 Bernoulli polynomials of the second kind

The historical terminology is not uniform. We use the Roman–MathWorld normalization, also called Cauchy polynomials of the first kind.

Definition 29.9. Define $b_n(x)$ by

$$\frac{t(1+t)^x}{\log(1+t)} = \sum_{n \geq 0} b_n(x) \frac{t^n}{n!}. \quad (29.17)$$

Theorem 29.10 (Integral representation and finite difference). For $n \geq 0$,

$$b_n(x) = \int_0^1 x + u^n \, du, \quad (29.18)$$

$$b_n(x+1) - b_n(x) = nb_{n-1}(x) \quad (n \geq 1). \quad (29.19)$$

Proof. Integrate the falling-factorial binomial expansion:

$$\int_0^1 (1+t)^{x+u} \, du = (1+t)^x \frac{t}{\log(1+t)}.$$

Coefficient comparison proves the integral. Replacing x by $x+1$ in the EGF and subtracting multiplies it by t , proving the difference law. $\square$

Remark 29.11. The EGF has Sheffer form with $A(t) = t/\log(1+t)$ and $B(t) = \log(1+t)$. Hence the lowering operator is the forward difference $e^D - 1$. The first numbers $b_n = b_n(0)$ are $1, 1/2, -1/6, 1/4, -19/30, 9/4, \dots$

29.5 A modified Bernoulli normalization

Define the even ordinary-series coefficients $\beta_{2m}^\sharp$ by

$$\frac{1}{2} \log \left(\frac{e^{x/2} - e^{-x/2}}{x/2} \right) = \sum_{m \geq 0} \beta_{2m}^\sharp x^{2m}, \quad \beta_0^\sharp = \frac{1}{2} \log 2. \quad (29.20)$$

Proposition 29.12. For even $n \geq 2$,

$$\beta_n^\sharp = \frac{\beta_n}{2n^2\Gamma(n)}, \quad (29.21)$$

and odd coefficients vanish. In particular, $\beta_2^\sharp = 1/48$, $\beta_4^\sharp = -1/5760$, and $\beta_6^\sharp = 1/362880$.

Proof. Differentiate the defining series:

$$\frac{1}{4} \coth \frac{x}{2} - \frac{1}{2x} = \frac{1}{2x} \left(\frac{x}{2} \coth \frac{x}{2} - 1 \right).$$

The Bernoulli generating function implies

$$\frac{x}{2} \coth \frac{x}{2} = \sum_{m \geq 0} \beta_{2m} \frac{x^{2m}}{(2m)!}.$$

Integrating termwise gives $\beta_{2m}^\sharp = \beta_{2m}/(4m(2m)!)$, which is the displayed formula. $\square$

29.6 Euler–Maclaurin summation

Put $P_r(x) = \beta_r(\{x\})/r!$, with $\{x\}$ the fractional part.

Theorem 29.13 (Euler–Maclaurin with remainder). Let $a < b$ be integers and $f \in C^{2m}([a, b])$. Then

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(x) \, dx + \frac{f(a) + f(b)}{2} \\ &\quad + \sum_{r=1}^m \frac{\beta_{2r}}{(2r)!} (f^{(2r-1)}(b) - f^{(2r-1)}(a)) + R_m, \end{aligned} \quad (29.22)$$

where

$$R_m = -\frac{1}{(2m)!} \int_a^b \beta_{2m}(\{x\}) f^{(2m)}(x) \, dx \quad (29.23)$$

and

$$|R_m| \leq \frac{2\zeta(2m)}{(2\pi)^{2m}} \int_a^b |f^{(2m)}(x)| \, dx. \quad (29.24)$$

Proof. On $[k, k+1]$, integration by parts gives

$$\int_k^{k+1} \left(x - k - \frac{1}{2} \right) f'(x) \, dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x) \, dx.$$

Summing yields the trapezoidal term plus $\int_a^b P_1 f'$. Integrate by parts repeatedly, using $P'_r = P_{r-1}$ away from integers and continuity of P_r at integers for $r \geq 2$. Boundary values are $\beta_r/r!$; odd Bernoulli numbers beyond β_1 vanish. The surviving boundary terms and final integral are exactly (29.22) and (29.23).

The periodic Bernoulli function has absolutely convergent Fourier series

$$\frac{\beta_{2m}(\{x\})}{(2m)!} = (-1)^{m+1} \frac{2}{(2\pi)^{2m}} \sum_{q \geq 1} \frac{\cos(2\pi qx)}{q^{2m}}.$$

Taking absolute values gives the supremum bound used in (29.24). $\square$

Corollary 29.14 (Stirling’s expansion). *For fixed M ,*

$$\log(n!) = \left(n + \frac{1}{2}\right) \log n - n + \frac{1}{2} \log(2\pi) + \sum_{r=1}^{M-1} \frac{\beta_{2r}}{2r(2r-1)n^{2r-1}} + O(n^{-(2M-1)}). \quad (29.25)$$

Proof. Apply Euler–Maclaurin to $f(x) = \log x$. Its derivatives give the displayed inverse powers. The terms at the lower endpoint and the limiting remainder combine into a constant, evaluated as $\frac{1}{2} \log(2\pi)$ by Wallis’ product. The remainder bound one order farther gives the stated error. $\square$

Chapter 30

Pochhammer symbols, polygamma functions, and Barnes' G -function

30.1 Pochhammer algebra

Definition 30.1. The rising Pochhammer symbol is

$$(a)_n = a^{\overline{n}} = a(a+1) \cdots (a+n-1), \quad (a)_0 = 1. \quad (30.1)$$

Theorem 30.2 (Gamma quotient and binomial series). *Where defined meromorphically,*

$$(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}, \quad (30.2)$$

and, formally in z and analytically for $|z| < 1$,

$$(1-z)^{-a} = \sum_{n \geq 0} \frac{(a)_n}{n!} z^n. \quad (30.3)$$

Proof. Iterate $\Gamma(w+1) = w\Gamma(w)$ to obtain the quotient. For the series, let $F = \sum (a)_n z^n / n!$. The recurrence $(a)_{n+1} = (a+n)(a)_n$ gives $(1-z)F' = aF$ and $F(0) = 1$, whose unique formal solution is $(1-z)^{-a}$. The ratio test gives analytic convergence in the unit disk. $\square$

Corollary 30.3 (Chu–Vandermonde).

$$\frac{(a+b)_n}{n!} = \sum_{k=0}^n \frac{(a)_k}{k!} \frac{(b)_{n-k}}{(n-k)!}. \quad (30.4)$$

Proof. Multiply the binomial series for exponents a and b and compare with the series for exponent $a+b$. $\square$

For $r \geq 1$, define shifted generalized harmonic sums

$$H_n^{(r)}(a) = \sum_{j=0}^{n-1} (a+j)^{-r}, \quad H_n(a) = H_n^{(1)}(a). \quad (30.5)$$

Theorem 30.4 (All parameter derivatives). *For $m \geq 1$ away from the apparent poles,*

$$\frac{\partial^m}{\partial a^m} (a)_n = (a)_n \mathcal{B}_m \left(H_n(a), -1! H_n^{(2)}(a), 2! H_n^{(3)}(a), \dots, (-1)^{m-1} (m-1)! H_n^{(m)}(a) \right). \quad (30.6)$$

For $m = 0$ the corresponding statement is the identity $(a)_n = (a)_n \mathcal{B}_0$, with $\mathcal{B}_0 = 1$. The right side extends polynomially across all removable singularities.

Proof. Write $L(a) = \log(a)_n$. Then $L^{(r)}(a) = (-1)^{r-1} (r-1)! H_n^{(r)}(a)$. Apply the exponential case of Faà di Bruno to $e^{L(a)}$. Since the left side is a polynomial in a , agreement off the zeros supplies the polynomial continuation. $\square$

30.2 Digamma and polygamma series

Define $\psi = \Gamma'/\Gamma$ and $\psi^{(m)} = D^m\psi$.

Theorem 30.5 (Partial fractions). *Away from nonpositive integers,*

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+z} \right), \quad (30.7)$$

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} (k+z)^{-m-1} \quad (m \geq 1). \quad (30.8)$$

Proof. Logarithmically differentiate Weierstrass' product

$$\frac{1}{\Gamma(z)} = z e^{\gamma z} \prod_{k \geq 1} \left(1 + \frac{z}{k} \right) e^{-z/k}.$$

Normal convergence on compact sets away from the poles justifies the first series and every further derivative. $\square$

Corollary 30.6 (Recurrences and special values).

$$\psi(z+1) = \psi(z) + z^{-1}, \quad (30.9)$$

$$\psi^{(m)}(z+1) = \psi^{(m)}(z) + (-1)^m m! z^{-m-1}, \quad (30.10)$$

$$\psi(n) = H_{n-1} - \gamma, \quad (30.11)$$

$$\psi^{(m)}(1) = (-1)^{m+1} m! \zeta(m+1). \quad (30.12)$$

Proof. Subtract the partial-fraction series at consecutive arguments; the series telescopes. Iterate from $\psi(1) = -\gamma$, and set $z = 1$ in the polygamma series. $\square$

Theorem 30.7 (Reflection). *For $z \notin \mathbb{Z}$,*

$$\psi(1-z) - \psi(z) = \pi \cot(\pi z), \quad (30.13)$$

$$(-1)^m \psi^{(m)}(1-z) - \psi^{(m)}(z) = \pi \frac{d^m}{dz^m} \cot(\pi z). \quad (30.14)$$

Proof. Multiplying the Weierstrass products for $1/\Gamma(1+z)$ and $1/\Gamma(1-z)$ and using Euler's sine product gives $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$. Logarithmic differentiation proves the first formula; differentiating it m more times proves the second. $\square$

Theorem 30.8 (Gauss multiplication). *For $q \in \mathbb{N}_{>0}$,*

$$\Gamma(qz) = (2\pi)^{(1-q)/2} q^{qz-1/2} \prod_{r=0}^{q-1} \Gamma\left(z + \frac{r}{q}\right). \quad (30.15)$$

Hence

$$\sum_{r=0}^{q-1} \psi\left(z + \frac{r}{q}\right) = q\psi(qz) - q \log q, \quad (30.16)$$

$$\sum_{r=0}^{q-1} \psi^{(m)}\left(z + \frac{r}{q}\right) = q^{m+1} \psi^{(m)}(qz). \quad (30.17)$$

Proof. Define

$$Q(z) = \frac{q^{qz-1/2} \prod_{r=0}^{q-1} \Gamma(z + r/q)}{\Gamma(qz)}.$$

The gamma recurrence shows $Q(z + 1/q) = Q(z)$. Stirling's formula in a right half-plane gives $Q(z) \rightarrow (2\pi)^{(q-1)/2}$ as $\Re z \rightarrow +\infty$ in a fixed horizontal strip. Periodicity therefore makes Q equal to that constant; meromorphic continuation extends the identity throughout its domain. Solving for $\Gamma(qz)$ proves (30.15). Logarithmic differentiation gives the digamma identity and further differentiation gives the polygamma identity. $\square$

Corollary 30.9. *For every $m \geq 0$,*

$$\frac{\partial^{m+1}}{\partial a^{m+1}} \log(a)_n = \psi^{(m)}(a+n) - \psi^{(m)}(a) = (-1)^m m! H_n^{(m+1)}(a). \quad (30.18)$$

Proof. Logarithmically differentiate the gamma quotient and compare with direct differentiation of the finite product. $\square$

30.3 Barnes' function

Theorem 30.10 (Canonical product). *The normally convergent product*

$$\begin{aligned} G(z+1) &= (2\pi)^{z/2} \exp\left(-\frac{z(z+1)}{2} - \frac{\gamma z^2}{2}\right) \\ &\quad \times \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right)^k \exp\left(-z + \frac{z^2}{2k}\right) \end{aligned} \quad (30.19)$$

defines an entire function satisfying

$$G(1) = 1, \quad G(z+1) = \Gamma(z)G(z). \quad (30.20)$$

Consequently $G(n) = 1!2! \cdots (n-2)!$ for $n \geq 2$.

Proof. The logarithm of the k th factor is $k \log(1 + z/k) - z + z^2/(2k) = z^3/(3k^2) + O_K(k^{-3})$ on a compact set K , so the product converges normally and defines an entire function with the prescribed integral-multiplicity zeros. At $z = 0$ it equals 1. Divide the finite products at z and $z - 1$, rearrange before taking the limit, and use Weierstrass' gamma product; all quadratic convergence factors cancel and the limit is $\Gamma(z)$. Normal convergence justifies passage to the limit, proving the recurrence. Iteration gives the integer values. $\square$

Theorem 30.11 (Alexeiewsky identity). *With compatible logarithm branches,*

$$\log G(z+1) = \frac{z}{2} \log(2\pi) - \frac{z(z+1)}{2} + z \log \Gamma(z+1) - \int_0^z \log \Gamma(t+1) dt. \quad (30.21)$$

Proof. Logarithmic differentiation of the product gives

$$\frac{G'(z+1)}{G(z+1)} = \frac{1}{2} \log(2\pi) - z - \frac{1}{2} + z\psi(z+1),$$

where the simplification uses the digamma partial fractions. The derivative of the right side of (30.21) is identical, and both sides vanish at $z = 0$. $\square$

Corollary 30.12. *For $n \geq 1$,*

$$\prod_{k=1}^n k^k = \frac{(n!)^n}{G(n+1)}. \quad (30.22)$$

Proof. In $(n!)^n/G(n+1)$, the exponent of k is $n - (n - k) = k$. □

Theorem 30.13 (Barnes asymptotics). As $z \rightarrow \infty$ in $|\arg z| \leq \pi - \delta$,

$$\begin{aligned} \log G(z+1) \sim & \frac{1}{4}z^2 + z \log \Gamma(z+1) - \left(\frac{1}{2}z(z+1) + \frac{1}{12} \right) \log z - \log A \\ & + \sum_{k \geq 1} \frac{\beta_{2k+2}}{2k(2k+1)(2k+2)z^{2k}}, \end{aligned} \quad (30.23)$$

where A is the Glaisher–Kinkelin constant. Equivalently,

$$\log A = \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n k \log k - \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{12} \right) \log n + \frac{n^2}{4} \right\}. \quad (30.24)$$

Proof. Insert Stirling's Euler–Maclaurin expansion for $\log \Gamma(t+1)$ into (30.21) and integrate termwise. The integrated Bernoulli terms, after shifting the index, are exactly the displayed inverse even powers; the integrated remainder has the next Poincaré order uniformly in each closed sector. The remaining integration constant is independent of z . Evaluate at positive integers using $\log G(n+1) = \sum_{j=1}^{n-1} \log(j!)$ and summation by parts; comparison with (30.24) identifies it as $-\log A$. This is the standard DLMF normalization of Barnes' G -function [16]. □

Remark 30.14. Polynomial identities such as the Pochhammer derivative formula are formal and require no convergence. The partial-fraction series and Barnes product are convergent analytic identities, while (30.23) is a Poincaré expansion. Keeping these three meanings of “equals” separate is essential in a reliable formula compendium.

Part IX

Catalan Refinements and Singularity Transfer

Chapter 31

Narayana numbers, polygon dissections, associahedra, and the Tamari order

The earlier Lagrange chapter derived the Catalan and Fuss–Catalan formulas. The Wikipedia and MathWorld dossiers add refinements and geometric models. We fix the convention that $\mathcal{K}_N$ is the associahedron attached to a convex N -gon; hence $\dim \mathcal{K}_N = N - 3$. This eliminates the two-unit index shifts common in sources that index instead by dimension or number of binary-tree leaves.

31.1 Dyck paths and reflection

A Dyck path of semilength n uses n up-steps $U = (1, 1)$ and n down-steps $D = (1, -1)$ and never falls below height 0.

Theorem 31.1 (Catalan formula). *The number of Dyck paths of semilength n is*

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (31.1)$$

Proof. Among all $\binom{2n}{n}$ balanced paths, reflect the prefix through the first step that reaches height -1 . This is a bijection from bad paths to words with $n+1$ up-steps and $n-1$ down-steps, counted by $\binom{2n}{n-1}$. Subtraction and simplification prove the formula. $\square$

Theorem 31.2 (First return). *The generating function $C(z) = \sum C_n z^n$ satisfies*

$$C(z) = 1 + zC(z)^2, \quad C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}. \quad (31.2)$$

Proof. Every nonempty path has a unique decomposition $UPDQ$, where D is the first return to height zero and P, Q are Dyck paths. This gives the functional equation. Of the two quadratic roots, only the displayed branch has constant term 1. $\square$

31.2 Narayana refinement

Let $N(n, k)$ count Dyck paths with k peaks, where a peak is an occurrence of UD , and let $C(z, u)$ mark semilength by z and peaks by u .

Theorem 31.3 (Narayana formula). *For $1 \leq k \leq n$,*

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \quad (31.3)$$

and

$$C(z, u) = 1 + zC(z, u)(C(z, u) + u - 1). \quad (31.4)$$

Consequently $\sum_k N(n, k) = C_n$ and $N(n, k) = N(n, n + 1 - k)$.

Proof. Put $H = C - 1$. In the first-return decomposition $UPDQ$, the initial arch is a peak exactly when P is empty. Hence $H = z(uC + (C - 1)C) = z(1 + H)(u + H)$. Lagrange inversion gives

$$[z^n u^k]H = \frac{1}{n} [w^{n-1} u^k] (1 + w)^n (u + w)^n = \frac{1}{n} \binom{n}{k-1} \binom{n}{k}.$$

Setting $u = 1$ gives the Catalan equation, and the closed formula is symmetric under $k \leftrightarrow n + 1 - k$. $\square$

Corollary 31.4 (Peak mean and variance). *For a uniformly random Dyck path of semilength $n \geq 2$, the number K_n of peaks satisfies*

$$\mathbb{E}K_n = \frac{n+1}{2}, \quad \text{Var}(K_n) = \frac{n^2 - 1}{4(2n - 1)}. \quad (31.5)$$

Proof. Symmetry gives the mean. Differentiate the Narayana polynomial at $u = 1$ and use Vandermonde's convolution:

$$\begin{aligned} \sum_k k \binom{n}{k} \binom{n}{k-1} &= n \binom{2n-1}{n}, \\ \sum_k k(k-1) \binom{n}{k} \binom{n}{k-1} &= n(n-1) \binom{2n-2}{n-1}. \end{aligned}$$

Divide by nC_n and convert the second factorial moment to the variance. $\square$

31.3 Explicit Catalan bijections

Theorem 31.5 (Binary trees and parenthesizations). *Dyck paths of semilength n , rooted plane full binary trees with n internal vertices, and full parenthesizations of $n + 1$ ordered factors are in mutually explicit bijection.*

Proof. Map the empty path to a leaf. Recursively map $UPDQ$ to an internal root whose left and right subtrees are the images of P, Q . The construction is reversible. A binary tree becomes a parenthesization by labelling its leaves from left to right and interpreting each internal node as the product of its two children; the final multiplication recovers the root split. $\square$

Theorem 31.6 (Triangulations). *Triangulations of a convex $(n + 2)$ -gon are counted by C_n .*

Proof. Fix a root boundary edge. The planar dual of a triangulation has one vertex per triangle and one edge across each internal diagonal. Attach a leaf across every nonroot boundary edge and root at the triangle incident with the root edge. Each triangle has two remaining sides, so the result is a full plane binary tree. Conversely, glue a triangle at each internal vertex along its parent side. The constructions are inverse, and there are n triangles. $\square$

Theorem 31.7 (Noncrossing partitions). *Noncrossing partitions of $[n]$ in circular order are counted by C_n .*

Proof. Let a_n count them. Decompose a partition at the block containing 1. The gaps between successive elements of that block carry independent noncrossing partitions. If the block has $r + 1$ elements, the gap sizes sum to $n - r - 1$. Therefore the OGF $A(z) = \sum a_n z^n$ satisfies

$$A - 1 = \sum_{r \geq 0} z^{r+1} A^{r+1} = \frac{zA}{1 - zA},$$

which rearranges to $A = 1 + zA^2$. The decomposition is reversible, so this is also a recursive bijection. $\square$

31.4 Dissections and the associahedron

Let $D(N, d)$ count sets of d pairwise noncrossing diagonals of a convex N -gon.

Theorem 31.8 (Kirkman–Cayley dissection formula). *For $0 \leq d \leq N - 3$,*

$$D(N, d) = \frac{1}{d+1} \binom{N+d-1}{d} \binom{N-3}{d}. \quad (31.6)$$

Proof. Root one boundary edge and take the planar dual of a dissection. Its $I = d + 1$ regions become internal vertices and its $L = N - 1$ other boundary edges become leaves. This is a rooted plane tree with no unary internal vertices, and the construction is reversible.

In preorder, record outdegree minus one. Internal entries are positive, leaves contribute -1 , and the total is -1 . Choose the I internal positions in $\binom{I+L}{I}$ ways and their positive values, summing to $L - 1$, in $\binom{L-2}{I-1}$ ways. The cycle lemma says exactly one of the $I + L$ cyclic shifts has nonnegative proper partial sums and hence encodes a plane tree. Thus the number is

$$\frac{1}{I+L} \binom{I+L}{I} \binom{L-2}{I-1} = \frac{1}{I} \binom{L+I-1}{I-1} \binom{L-2}{I-1},$$

which becomes (31.6) after substitution. $\square$

Corollary 31.9 (Associahedral face vector). *A j -face of $\mathcal{K}_N$ corresponds to $N - 3 - j$ fixed noncrossing diagonals, and*

$$f_j(\mathcal{K}_N) = \frac{1}{N-2-j} \binom{2N-4-j}{N-3-j} \binom{N-3}{j}. \quad (31.7)$$

Thus $f_0 = C_{N-2}$, $f_{N-4} = N(N-3)/2$, and $f_{N-3} = 1$.

Proof. A face is a partial triangulation, equivalently a set of fixed noncrossing diagonals. Its dimension is the number of diagonals still free, so substitute $d = N - 3 - j$ in (31.6) and simplify. $\square$

31.5 The Tamari order

Orient the elementary rotation by

$$((AB)C) \longrightarrow (A(BC)). \quad (31.8)$$

Theorem 31.10 (The rotation relation is a partial order). *Reachability under (31.8) is antisymmetric and therefore defines the Tamari partial order.*

Proof. Label the leaves $1, \dots, N$ from left to right and let $d_T(i)$ be the depth of leaf i . Define the integer potential

$$\Phi(T) = \sum_{i=1}^N 2^i d_T(i).$$

Under a rotation $((AB)C) \rightarrow (A(BC))$, the depths of leaves in A decrease by one, those in C increase by one, and those in B are unchanged. Hence

$$\Phi(A(BC)) - \Phi((AB)C) = \sum_{i \in C} 2^i - \sum_{i \in A} 2^i > 0.$$

Indeed every index in C is larger than every index in A , and the smallest power of two occurring in the first sum exceeds the sum of all preceding powers. Thus the potential strictly increases along each directed rotation, so no directed cycle exists. Reachability is reflexive and transitive by definition, and acyclicity gives antisymmetry. $\square$

Theorem 31.11 (Raney numbers). *If $T = 1 + zT^p$, then for integers $p, r \geq 1$ and $n \geq 0$,*

$$[z^n]T(z)^r = \frac{r}{pn+r} \binom{pn+r}{n}. \quad (31.9)$$

Proof. The case $n = 0$ is immediate. For $n \geq 1$, put $Y = T - 1 = z(1 + Y)^p$. Lagrange–Bürmann applied to $(1 + Y)^r$ gives

$$[z^n]T^r = \frac{r}{n} [w^{n-1}](1 + w)^{pn+r-1} = \frac{r}{pn+r} \binom{pn+r}{n}.$$

□

Chapter 32

Darboux's finite formula and transfer from singularities

The MathWorld entry emphasizes an exact finite integration identity, while analytic combinatorics uses Darboux's principle to transfer local singular expansions to coefficient asymptotics. Both are proved here; the latter is stated in a form sufficient for the Catalan and inverse-function examples [7].

32.1 Finite Darboux identity

Theorem 32.1 (Darboux formula). *Let f have $n + 1$ continuous derivatives on the segment from a to z , and let ϕ be a polynomial of degree at most n . Then*

$$\begin{aligned}\phi^{(n)}(0)(f(z) - f(a)) &= \sum_{m=1}^n (-1)^{m-1} (z-a)^m \left(\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right) \\ &\quad + (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}(a + t(z-a)) dt.\end{aligned}\tag{32.1}$$

Proof. Put $d = z - a$, $F_m(t) = f^{(m)}(a + td)$, and

$$S(t) = \sum_{m=1}^n (-1)^{m-1} d^m \phi^{(n-m)}(t) F_m(t).$$

Differentiate and reindex the terms in which the derivative hits F_m . Every interior term cancels, leaving

$$S'(t) = d\phi^{(n)}(0)F_1(t) + (-1)^{n-1}d^{n+1}\phi(t)F_{n+1}(t).$$

Integrate from 0 to 1, move the last integral to the other side, and expand $S(1) - S(0)$. $\square$

Corollary 32.2 (Taylor integral remainder).

$$f(z) = \sum_{m=0}^n \frac{f^{(m)}(a)}{m!} (z-a)^m + \frac{(z-a)^{n+1}}{n!} \int_0^1 (1-t)^n f^{(n+1)}(a + t(z-a)) dt.\tag{32.2}$$

Proof. Take $\phi(t) = (1-t)^n$ in the theorem. All derivatives below order n vanish at $t = 1$, and $\phi^{(n-m)}(0) = (-1)^{n-m} n! / m!$. $\square$

32.2 Algebraic singularities

Lemma 32.3 (Exact binomial coefficient). *For $R \neq 0$,*

$$[z^n](1 - z/R)^{-\alpha} = R^{-n} \frac{\Gamma(n + \alpha)}{\Gamma(\alpha)\Gamma(n + 1)}, \quad (32.3)$$

where $1/\Gamma(\alpha)$ supplies continuation at nonpositive integers. For fixed admissible α ,

$$[z^n](1 - z/R)^{-\alpha} = \frac{R^{-n}n^{\alpha-1}}{\Gamma(\alpha)} \left(1 + \frac{\alpha(\alpha-1)}{2n} + O(n^{-2}) \right). \quad (32.4)$$

Proof. The exact formula is the generalized binomial series. Subtract the Stirling expansions of $\log \Gamma(n + \alpha)$ and $\log \Gamma(n + 1)$ and exponentiate to obtain the ratio expansion. $\square$

Theorem 32.4 (Darboux transfer). *Let $0 < R < \rho$, let h be analytic in $|z| < \rho$, and fix $J \geq 1$. Then*

$$\begin{aligned} [z^n]h(z)(1 - z/R)^{-\alpha} &= R^{-n} \sum_{j=0}^{J-1} \frac{(-R)^j h^{(j)}(R)}{j!} \frac{\Gamma(n + \alpha - j)}{\Gamma(\alpha - j)\Gamma(n + 1)} \\ &\quad + O\left(R^{-n}n^{\Re\alpha-J-1}\right). \end{aligned} \quad (32.5)$$

In particular, if $h(R) \neq 0$,

$$[z^n]h(z)(1 - z/R)^{-\alpha} \sim \frac{h(R)}{\Gamma(\alpha)} R^{-n}n^{\alpha-1}. \quad (32.6)$$

Proof. Taylor-expand $h(z) = \sum_{j < J} h^{(j)}(R)(z - R)^j/j! + (z - R)^J q(z)$. Since $(z - R)^j = (-R)^j(1 - z/R)^j$, the finite terms give the displayed gamma ratios. Choose r with $R < r < \rho$; Cauchy's estimate gives $[z^k]q = O(r^{-k})$. Convoluting this geometric bound with $[z^m](1 - z/R)^{J-\alpha} = O(R^{-m}(m+1)^{\Re\alpha-J-1})$ gives the remainder, because $R/r < 1$. The leading case follows from the gamma-ratio asymptotic. $\square$

Corollary 32.5 (Multiple dominant points). *If finitely many local singular models occur at points ρ_j of common modulus R and the remainder is analytic past that circle, their coefficient contributions add:*

$$[z^n]F(z) = \sum_j \frac{h_j(\rho_j)}{\Gamma(\alpha_j)} \rho_j^{-n} n^{\alpha_j-1} (1 + O(n^{-1})). \quad (32.7)$$

Complex conjugate points therefore produce periodic or oscillatory terms.

Proof. Subtract the finite singular models and apply Cauchy's estimate to the analytic remainder; then apply Theorem 32.4 term by term. $\square$

Proposition 32.6 (Logarithmic and meromorphic cases). *For $n \geq 1$,*

$$[z^n] \log \frac{1}{1 - z/R} = \frac{R^{-n}}{n}. \quad (32.8)$$

If a meromorphic function has principal parts $c_{j,m}(1 - z/\rho_j)^{-m}$ at its poles of smallest modulus, then its coefficients equal

$$\sum_{j,m} c_{j,m} \rho_j^{-n} \binom{n+m-1}{m-1} \quad (32.9)$$

up to an exponentially smaller contribution from the analytic remainder.

Proof. The logarithmic formula is its Taylor series. For poles, subtract all principal parts, use the integer binomial coefficient formula, and apply Cauchy's estimate to what remains. $\square$

32.3 Catalan asymptotics

Theorem 32.7. As $n \rightarrow \infty$,

$$C_n = \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} + O(n^{-3}) \right). \quad (32.10)$$

Proof. Use $C_n = (n+1)^{-1} \binom{2n}{n}$ and

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + \frac{1}{128n^2} + O(n^{-3}) \right),$$

which follows from Stirling's expansion. Multiply by $(n+1)^{-1} = n^{-1}(1 - n^{-1} + n^{-2} + O(n^{-3}))$. Alternatively, $C(z) = 2 - 2\sqrt{1-4z} + O(1-4z)$ near $z = 1/4$, and Darboux transfer with $\alpha = -1/2$ gives the leading term directly. $\square$

Remark 32.8 (Choosing the asymptotic method). Darboux transfer applies to fixed finite singularities. The Bell EGF $\exp(e^z - 1)$ is entire, so its coefficients require the moving saddle point and Lambert W developed earlier. Implicit algebraic functions are normally handled by Lagrange inversion followed by singularity transfer; multivariate systems require the characteristic determinant and often a saddle analysis.

Part X

Further Differential and Reversion Examples

Chapter 33

Higher product rules, quotients, and elementary inverse series

The five drafts contain several useful consequences of the two master composition theorems that do not belong to a single classical triangle. They are collected here rather than repeated in the Bell-polynomial and inversion chapters.

33.1 Products, reciprocals, and quotients

Theorem 33.1 (Multinomial Leibniz rule). *For functions $f_1, \dots, f_q$ possessing n derivatives,*

$$D^n \left(\prod_{r=1}^q f_r(x) \right) = \sum_{j_1 + \dots + j_q = n} \binom{n}{j_1, \dots, j_q} \prod_{r=1}^q f_r^{(j_r)}(x). \quad (33.1)$$

Proof. Taylor's formula, interpreted either analytically near x or formally in a new variable h , gives

$$\prod_{r=1}^q f_r(x+h) = \prod_{r=1}^q \left(\sum_{j_r \geq 0} f_r^{(j_r)}(x) \frac{h^{j_r}}{j_r!} \right).$$

The coefficient of h^n is the sum over $j_1 + \dots + j_q = n$ of $\prod_r f_r^{(j_r)}(x)/j_r!$. Multiplication by $n!$ proves (33.1). $\square$

Theorem 33.2 (Higher reciprocal and quotient rules). *If $g(x) \neq 0$, then*

$$\left(\frac{1}{g} \right)^{(n)} = \sum_{k=0}^n (-1)^k k! \frac{\mathcal{B}_{n,k}(g', g'', \dots, g^{(n-k+1)})}{g^{k+1}}. \quad (33.2)$$

Consequently,

$$D^n \left(\frac{f}{g} \right) = \sum_{r=0}^n \binom{n}{r} f^{(n-r)} \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}(g', g'', \dots, g^{(r-k+1)})}{g^{k+1}}. \quad (33.3)$$

For $n > 0$ the $k = 0$ term in (33.2) vanishes.

Proof. Apply Faà di Bruno's formula to $F(g(x))$ with $F(u) = u^{-1}$. Since $F^{(k)}(u) = (-1)^k k! u^{-k-1}$, the Bell-polynomial form gives (33.2). Formula (33.3) follows by applying the ordinary two-factor Leibniz rule to $f g^{-1}$ and substituting the reciprocal formula. $\square$

Corollary 33.3 (Trigonometric outer functions). *For $n \geq 0$,*

$$D^n \sin(g(x)) = \sum_{k=0}^n \sin\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}(g', g'', \dots), \quad (33.4)$$

$$D^n \cos(g(x)) = \sum_{k=0}^n \cos\left(g(x) + \frac{k\pi}{2}\right) \mathcal{B}_{n,k}(g', g'', \dots). \quad (33.5)$$

Proof. Use $D^k \sin u = \sin(u + k\pi/2)$ and $D^k \cos u = \cos(u + k\pi/2)$ in Faà di Bruno's formula. $\square$

33.2 Diagonal identities furnished by inverse functions

Proposition 33.4 (Inverse sine and tangent diagonals). *For $m \geq 0$,*

$$\arcsin y = \sum_{m \geq 0} \frac{1}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}, \quad (33.6)$$

$$[x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1} = \frac{1}{4^m} \binom{2m}{m}, \quad (33.7)$$

$$[x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1} = (-1)^m. \quad (33.8)$$

Proof. The binomial theorem gives $(1 - y^2)^{-1/2} = \sum_{m \geq 0} \binom{2m}{m} y^{2m} / 4^m$; integrating from 0 to y proves (33.6). Apply analytic Lagrange inversion to the inverse pair $y = \sin x$, $x = \arcsin y$:

$$[y^{2m+1}] \arcsin y = \frac{1}{2m+1} [x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1}.$$

Comparison with (33.6) proves (33.7). Similarly, $\arctan y = \sum_{m \geq 0} (-1)^m y^{2m+1} / (2m+1)$ follows by integrating $(1 + y^2)^{-1}$. Lagrange inversion for $y = \tan x$ then gives (33.8). $\square$

Proposition 33.5 (Exponential–logarithmic diagonals). *For $n \geq 1$,*

$$[x^{n-1}] \left(\frac{x}{e^x - 1} \right)^n = (-1)^{n-1}, \quad (33.9)$$

$$[x^{n-1}] \left(\frac{x}{\log(1+x)} \right)^n = \frac{1}{(n-1)!}. \quad (33.10)$$

Proof. The inverse of $e^x - 1$ is $\log(1 + y)$, whose coefficient of y^n is $(-1)^{n-1}/n$. Formula (21.4) therefore gives (33.9). Conversely, the inverse of $\log(1 + x)$ is $e^y - 1$, whose coefficient of y^n is $1/n!$; the same formula gives (33.10). $\square$

33.3 Expansion about a nonzero equilibrium

Theorem 33.6 (Offset Lagrange–Bürmann formula). *Let $u(t)$ be the unique formal solution of*

$$u = a + t\Phi(u), \quad \Phi(a) \neq 0. \quad (33.11)$$

Then for every formal H and $n \geq 1$,

$$[t^n] H(u(t)) = \frac{1}{n} [z^{n-1}] H'(a + z) \Phi(a + z)^n. \quad (33.12)$$

The same formula holds analytically in a sufficiently small neighbourhood when H and Φ are analytic.

Proof. Put $w = u - a$ and $\phi(w) = \Phi(a + w)$. Then $w = t\phi(w)$, so Theorem 21.1, applied to the outer series $K(w) = H(a + w)$, gives

$$[t^n]K(w(t)) = \frac{1}{n}[z^{n-1}]K'(z)\phi(z)^n,$$

which is (33.12). The analytic statement follows from the analytic inverse-function theorem exactly as in Theorem 21.2. $\square$

Chapter 34

Kepler's equation as a complete reversion example

A classical nonlinear equation with both a formal coefficient expansion and a globally convergent Fourier representation is

$$E = M + e \sin E, \quad 0 \leq e < 1. \quad (34.1)$$

Here M is the mean anomaly, E the eccentric anomaly, and e is treated mathematically as a parameter.

Theorem 34.1 (Lagrange and Fourier–Bessel forms). *The unique real solution of (34.1) satisfies*

$$E = M + \sum_{k \geq 1} \frac{e^k}{k!} \left(\frac{d}{dM} \right)^{k-1} \sin^k M, \quad (34.2)$$

$$= M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM), \quad (34.3)$$

where, for integral $n \geq 0$,

$$J_n(z) = \frac{1}{2\pi} \int_0^{2\pi} \cos(n\theta - z \sin \theta) d\theta. \quad (34.4)$$

The first formula is an identity of formal series in e ; for every fixed $0 \leq e < 1$, the second series converges absolutely and uniformly in real M .

Proof. Formula (34.2) is the shifted reversion formula (23.3) with $x = M$, $y = e$, and $f(x) = \sin x$.

For the Fourier form, the function $M(E) = E - e \sin E$ has derivative $1 - e \cos E \geq 1 - e > 0$. Hence it has a smooth inverse satisfying $E(M + 2\pi) = E(M) + 2\pi$, and $E(M) - M$ is a smooth odd 2π -periodic function. Write

$$E(M) - M = \sum_{n \geq 1} c_n \sin(nM).$$

Its sine coefficient, after changing variables from M to E , is

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^{2\pi} (E - M) \sin(nM) dM \\ &= \frac{e}{\pi} \int_0^{2\pi} \sin E \sin(n(E - e \sin E)) (1 - e \cos E) dE. \end{aligned}$$

Since

$$\frac{d}{dE} \cos(n(E - e \sin E)) = -n(1 - e \cos E) \sin(n(E - e \sin E)),$$

integration by parts, with vanishing boundary term, gives

$$c_n = \frac{e}{\pi n} \int_0^{2\pi} \cos E \cos(nE - ne \sin E) dE = \frac{e}{n} (J_{n-1}(ne) + J_{n+1}(ne)).$$

It remains to prove the recurrence used to simplify this expression. The integral over a full period of $\frac{d}{d\theta} \sin(n\theta - z \sin \theta)$ is zero; therefore

$$n \int_0^{2\pi} \cos(n\theta - z \sin \theta) d\theta = z \int_0^{2\pi} \cos \theta \cos(n\theta - z \sin \theta) d\theta.$$

The product-to-sum identity and (34.4) yield $J_{n-1}(z) + J_{n+1}(z) = 2nJ_n(z)/z$, with the value at $z = 0$ understood by continuity. Thus $c_n = 2J_n(ne)/n$, proving (34.3). Finally, the inverse $E(M)$ is smooth for $e < 1$; repeated integration by parts makes its Fourier coefficients decay faster than every fixed inverse power of n , which proves absolute and uniform convergence. $\square$

Remark 34.2. The two expansions organize the same solution in different coordinates. The Lagrange series is local in the perturbation parameter e and exposes finite derivative polynomials at every order. The Fourier–Bessel series is global in the angular variable M and diagonalizes the periodic translation symmetry. Their equality is an instructive example of coefficient calculus as a change of basis rather than a collection of unrelated formulas.

Part XI

Algorithms, Normalization, and Reproducible Certificates

Chapter 35

Exact coefficient algorithms and consistency checks

A formula compendium is reliable only when its normalizations can be checked and its coefficients reproduced. This chapter merges the algorithmic and proof-audit appendices of the drafts. Every recurrence is triangular and can therefore be implemented over exact integers, rationals, finite fields, or symbolic coefficient rings.

35.1 Normalization dictionary

Ordinary versus exponential coefficients

If

$$f(z) = \sum_{n \geq 0} \alpha_n z^n = \sum_{n \geq 0} f_n \frac{z^n}{n!},$$

then $f_n = n! \alpha_n$. With the conventions of this monograph,

$$\widehat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (35.1)$$

Signed first-kind Stirling numbers are $s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}$; inverse-matrix identities always use the signed array.

Proof of (35.1). The exponential definition gives

$$\frac{1}{k!} \left(\sum_{j \geq 1} x_j t^j \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots) \frac{t^n}{n!}.$$

The coefficient on the left is $\widehat{\mathcal{B}}_{n,k}(x)/k!$ in ordinary normalization. Comparison proves the conversion. $\square$

Object	Defining expansion	Primary role
$s(n, k)$	$x^{\underline{n}} = \sum_k s(n, k)x^k$	inverse factorial basis
$\begin{bmatrix} n \\ k \end{bmatrix}$	$x^{\overline{n}} = \sum_k \begin{bmatrix} n \\ k \end{bmatrix} x^k$	permutation cycles
$\{n\}_k$	$x^n = \sum_k \{n\}_k x^{\underline{k}}$	set partitions
$\mathcal{B}_{n,k}$	exponential partial Bell polynomial	derivatives and labelled composition
$\widehat{\mathcal{B}}_{n,k}$	ordinary partial Bell polynomial	ordinary series composition
B_n	$\sum_k \{n\}_k$	unweighted partitions
β_n	$t/(e^t - 1) = \sum \beta_n t^n/n!$	power sums and Euler–Maclaurin

35.2 Elementary series operations

Let $A(z) = \sum a_n z^n$, $B(z) = \sum b_n z^n$, and $C(z) = \sum c_n z^n$.

Algorithm 35.1 (Product, reciprocal, and quotient). For $C = AB$,

$$c_n = \sum_{j=0}^n a_j b_{n-j}. \quad (35.2)$$

If a_0 is invertible and $C = A^{-1}$,

$$c_0 = a_0^{-1}, \quad c_n = -a_0^{-1} \sum_{j=1}^n a_j c_{n-j}. \quad (35.3)$$

For $C = B/A$,

$$c_n = a_0^{-1} \left(b_n - \sum_{j=1}^n a_j c_{n-j} \right). \quad (35.4)$$

Proof. The product formula is Cauchy's rule. Equate coefficients in $AC = 1$ or $AC = B$ and isolate the term $a_0 c_n$. $\square$

Algorithm 35.2 (Logarithm, exponential, and powers). Assume $a_0 = 1$. If $L = \log A = \sum_{n \geq 1} \ell_n z^n$, then

$$\ell_n = a_n - \frac{1}{n} \sum_{j=1}^{n-1} j \ell_j a_{n-j}. \quad (35.5)$$

If $A = e^L$, then

$$a_n = \frac{1}{n} \sum_{j=1}^n j \ell_j a_{n-j}. \quad (35.6)$$

If $C = A^\alpha$, then

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{j=1}^n ((\alpha + 1)j - n) a_j c_{n-j}. \quad (35.7)$$

Proof. Compare coefficients in $A' = AL'$ to obtain the first two recurrences. For powers, compare degree $n - 1$ in $AC' = \alpha A'C$ and isolate nc_n . $\square$

Algorithm 35.3 (Differentiation and integration). If $C = A'$, then $c_n = (n + 1)a_{n+1}$. If $C(z) = c_0 + \int_0^z A(t) dt$, then $c_{n+1} = a_n/(n + 1)$.

35.3 Composition and reversion

Algorithm 35.4 (Composition power table). Assume $B(0) = 0$ and set $p_{k,n} = [z^n]B(z)^k$. Initialize $p_{0,0} = 1$, $p_{0,n} = 0$ for $n > 0$, and use

$$p_{k,n} = \sum_{j=1}^n b_j p_{k-1,n-j}, \quad [z^n]A(B(z)) = \sum_{k=0}^n a_k p_{k,n}. \quad (35.8)$$

Proof. The first recurrence is $B^k = B B^{k-1}$; the second is $A(B) = \sum_k a_k B^k$. Since $B(0) = 0$, only finitely many powers contribute to each degree. $\square$

Algorithm 35.5 (Lagrange reversion). If $F(z) = a_1 z + a_2 z^2 + \dots$ with invertible a_1 , then its inverse $G(w) = \sum_{n \geq 1} g_n w^n$ has

$$g_n = \frac{1}{n} [z^{n-1}] \left(\frac{z}{F(z)} \right)^n. \quad (35.9)$$

Proof. This is Lagrange–Bürmann with outer function $H(z) = z$. The coefficient g_n depends only on $a_1, \dots, a_n$, so every truncation is exact through its declared order. $\square$

Algorithm 35.6 (Newton reversion). If $F(G_m(w)) \equiv w \pmod{w^m}$, define

$$G_{2m} = G_m - \frac{F(G_m) - w}{F'(G_m)} \pmod{w^{2m}}. \quad (35.10)$$

Then $F(G_{2m}) \equiv w \pmod{w^{2m}}$.

Proof. Put $E = F(G_m) - w = O(w^m)$ and $\Delta = -E/F'(G_m) = O(w^m)$. Taylor's formula in the formal ring gives $F(G_m + \Delta) = F(G_m) + F'(G_m)\Delta + O(\Delta^2) = w + O(w^{2m})$. The denominator is invertible because its constant term is a_1 . $\square$

Remark 35.7 (Arithmetic complexity). The triangular product, reciprocal, exponential, and logarithm recurrences use $O(N^2)$ scalar operations through degree N ; a direct composition table uses $O(N^3)$. Faster multiplication and Newton iteration reduce asymptotic cost, but the triangular forms are preferable for symbolic audits because every coefficient carries a short dependency certificate.

35.4 Small exact cross-checks

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0
2	0	1	1	0	0	0	0	2	0	1	1	0	0	0	0
3	0	2	3	1	0	0	0	3	0	1	3	1	0	0	0
4	0	6	11	6	1	0	0	4	0	1	7	6	1	0	0
5	0	24	50	35	10	1	0	5	0	1	15	25	10	1	0
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

(*)

The left table contains the unsigned first-kind Stirling numbers; its n th row sums to $n!$. The right table contains the second-kind Stirling numbers, whose row sums through $n = 6$ are

$$1, 1, 2, 5, 15, 52, 203.$$

After the first-kind array is signed, its product with the second-kind array must be the identity through this order.

Proposition 35.8 (Recurrence uniqueness). *If two triangular arrays have the same boundary data and obey the same recurrence from row $n - 1$ to row n , then they agree identically.*

Proof. Induct on the row number; every new entry is determined by already equal entries in the preceding row. $\square$

35.5 Deterministic and modular certificates

Theorem 35.9 (Polynomial grid certificate). *Let $P, Q \in K[x_1, \dots, x_d]$ and suppose $\deg_{x_i}(P - Q) \leq D_i$. If $P = Q$ on a Cartesian grid $S_1 \times \dots \times S_d$ with $D_i + 1$ distinct values in S_i , then $P = Q$ identically.*

Proof. For one variable, a nonzero degree- D polynomial has at most D roots. Induct on d , viewing $P - Q$ as a polynomial in x_d . At each grid point in the first $d - 1$ coordinates all its coefficient polynomials vanish; the induction hypothesis makes each coefficient identically zero. $\square$

Theorem 35.10 (Chinese-remainder certificate). *If an integer N has a proved bound $|N| < M/2$, where $M = p_1 \cdots p_s$ and the p_i are pairwise coprime, and if $N \equiv 0 \pmod{p_i}$ for every i , then $N = 0$.*

Proof. The congruences give $M \mid N$. The only multiple of M in $(-M/2, M/2)$ is zero. $\square$

Remark 35.11. Random evaluations are excellent error detectors but become proofs only after a degree bound (Schwartz–Zippel or a full deterministic grid) is supplied. Modular computations become exact certificates only after an integer-size bound is recorded. This distinction prevents numerical evidence from being silently promoted to an identity.

35.6 Certificate for the Bell-number constant

Put

$$R_n = \frac{B_n^{1/n} \log(n+1)}{n}. \quad (35.11)$$

The following entries are upper endpoints rounded upward to six decimals, computed from exact Bell integers and directed interval evaluation.

n	upper R_n	n	upper R_n	n	upper R_n
1	0.693148	19	0.740692	37	0.707939
2	0.776837	20	0.738200	38	0.706638
3	0.790177	21	0.735818	39	0.705373
4	0.791840	22	0.733538	40	0.704143
5	0.789788	23	0.731353	41	0.702946
6	0.786239	24	0.729257	42	0.701780
7	0.782127	25	0.727243	43	0.700644
8	0.777875	26	0.725307	44	0.699537
9	0.773672	27	0.723442	45	0.698457
10	0.769603	28	0.721646	46	0.697403
11	0.765707	29	0.719913	47	0.696375
12	0.761995	30	0.718240	48	0.695370
13	0.758468	31	0.716623	49	0.694388
14	0.755118	32	0.715059	50	0.693429
15	0.751938	33	0.713545	51	0.692491
16	0.748915	34	0.712079	52	0.691573
17	0.746040	35	0.710657	53	0.690675
18	0.743302	36	0.709278	54	0.689795

Proposition 35.12 (Finite Bell certificate). *The maximum in the table occurs at $n = 4$:*

$$R_4 = \frac{15^{1/4} \log 5}{4} = 0.7918392970 \dots < 0.792. \quad (35.12)$$

Together with the analytic estimate already proved for $n \geq 39$, this gives an overlapping certificate of $B_n < (0.792n / \log(n+1))^n$ for every $n \geq 1$.

Proof. Generate exact Bell integers by $B_0 = 1$ and $B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k$. Directed interval arithmetic for logarithm and exponentiation proves each upward-rounded entry. All are below 0.792, and exact interval comparison locates the maximum at $n = 4$. The table covers $1 \leq n \leq 54$ and the analytic estimate covers $n \geq 39$, leaving no gap. $\square$

35.7 Verification protocol

A robust workflow for a new identity is: normalize all signed/unsigned and ordinary/exponential conventions; derive the identity by coefficient extraction, a bijection, recurrence uniqueness, or incidence inversion; expand through at least fourth order, where the Faà di Bruno multiplicities 1, 4, 3, 6, 1 expose most missing factorials; and retain exact or bounded modular certificates for every computational step. The derivation is the proof; the other stages protect it against transcription and implementation errors.

Part XII

Source Concordance, Deduplication Record, and Verification

Chapter 36

How the five drafts were unified

The supplied `articles/` directory contains five substantial drafts, not five independent subjects. Their main theorems overlap, but each draft also contains material absent from the others. The merger therefore used theorem-level rather than paragraph-level deduplication. For every repeated topic, one statement and one complete proof were retained; genuinely different proofs were kept when they reveal different structures, such as a bijection versus a generating-function argument, or a formal-residue proof versus an analytic contour proof.

36.1 Draft-by-draft contribution map

Draft	Distinct strengths	Use in the unified article
<code>articles/combinatorial_coefficient_calculus_bundle/combinatorial_coefficient_calculus.tex</code>	The broadest proof-oriented core: restricted Stirling families, Bell-number interpretations and bounds, saddle methods, geometric Eulerian theory, normalized associahedral inversion, and Laplace asymptotics.	Used as the structural spine. Its collision-free notation and separation of formal from analytic claims were retained; duplicated introductions and ledgers were replaced by the present concordance.
<code>articles/Combinatorial_Formulae_and_Inversion_Theorems/Combinatorial_Formulae_and_Inversion_Theorems.tex</code>	Especially detailed change-of-basis proofs, probability interpretations, zeta and gamma series, two-parameter reversion, and a compact collection of reverse recurrences and boundary conventions.	Its nonduplicate derivations were reconciled with the Stirling, Bell, and shifted-reversion chapters. Equivalent displays were collapsed to one theorem with signed and unsigned conventions stated explicitly.

Continued on the next page

Draft	Distinct strengths	Use in the unified article
articles/Formulae_Compensum/Formulae_Compensum.tex	Incidence algebra, partition-lattice Möbius inversion, Sheffer sequences, exponential Riordan arrays, coordinate-free Faà di Bruno, recursive inverse derivatives, Good inversion, elementary inverse examples, and coefficient algorithms.	Supplies the main new material in Chapters 25, 27, 28, 33, and 35. Its multivariate determinant was rederived from the Jacobian so that row and column conventions cannot be confused.
articles/Formulae_Monograph_Package/Formulae_Monograph.tex	Strong normalization tables, low-order cross-checks, exact finite certificates, a Bell-number constant audit, and concise formula-family dependency maps.	Supplies the reproducibility material in Chapter 35. Numerical assertions are presented with exact recurrences and interval or modular certificate procedures rather than as unsupported decimal evidence.
articles/Formulae_Monograph_Source_and_PDF/Formulae_Monograph.tex	A polished synthesis of the composition and inversion chapters, including Fréchet derivatives, the delta-function interpretation of reversion, and Kepler's equation.	Its distinct higher-dimensional and physical examples are retained in Chapters 27 and 34; repeated Stirling, Bell, Eulerian, and Lagrange material was absorbed into the spine.

36.2 Deduplication rules

The merger applied the following rules consistently.

1. *Definitions are unique.* Every array or polynomial family has one canonical defining equation. Alternate normalizations appear only in an explicit conversion formula.
2. *A theorem is stated once.* Repeated formulas for Stirling transforms, Bell-polynomial composition, Faà di Bruno, and Lagrange–Bürmann were merged at the theorem level. Independent proofs are retained only when they expose different mechanisms.
3. *Boundary values are part of the theorem.* Empty sums, 0^0 in coefficient formulas, signed versus unsigned first-kind numbers, and the indexing of type- B Eulerian numbers are fixed globally.
4. *Formal and analytic assertions are separated.* Algebraic identities in formal series require no convergence; radii, branches, termwise differentiation, and asymptotic remainders are asserted only after suitable analytic hypotheses are supplied.
5. *Computations carry certificates.* A finite table, primality report, modular period, or numerical constant is not called a symbolic theorem. The recurrence, exact integer data, interval enclosure, or modular witness needed to reproduce it is recorded.

Chapter 37

Supplemental Wikipedia and MathWorld material

The encyclopedic files in `Wikipedia/` and `MathWorld/` were used as topic checklists and formula-discovery aids, not as substitutes for proofs. Their formulas were normalized to the notation of this monograph and proved from the common generating-function, residue, incidence-algebra, or analytic framework.

37.1 Wikipedia dossier

Files	Material checked	Integrated location
<code>Stirling_number.txt</code> , <code>Stirling_numbers_of_the_first_kind.txt</code> , <code>Stirling_numbers_of_the_second_kind.txt</code>	Factorial-basis definitions, cycle and partition interpretations, recurrences, inclusion-exclusion, generating functions, congruences, asymptotics, negative indices, and restricted variants.	Parts “Stirling Numbers, Lah Numbers, and Basis Changes” and “Coefficient Calculus and Labelled Structures.” Conflicting signs and edge conventions are resolved by the inverse-matrix definitions.
<code>Bell_number.txt</code> , <code>Bell_polynomials.txt</code>	Set partitions, Bell–Aitken triangle, Dobiński and Touchard formulas, congruences, saddle growth, partial and complete Bell polynomials, composition, moments, cumulants, and determinants.	Parts “Bell Numbers and Set Partitions” and “Bell Polynomials and Composition,” together with Chapter 26 for the signed complementary transform.
<code>Eulerian_number.txt</code>	Descent recurrences, Eulerian polynomials, Worpitzky interpolation, power sums, polylogarithms, real-rootedness, type B , second order, and geometric models.	Part “Eulerian Numbers and Descent Calculus.” The zero-position descent in type B is fixed before any recurrence is stated.

Continued on the next page

Files	Material checked	Integrated location
Faà di Bruno formula file (whose archive filename uses an escaped character)	Set-partition, multiplicity-vector, Bell-polynomial, and multivariate forms of the higher chain rule.	The original Faà di Bruno chapter and Chapter 27. The filename is rendered in standard spelling while all coefficient normalizations are derived.
Lagrange_inversion_theorem.txt, Lagrange_reversion_theorem.txt	Formal residues, analytic inversion, shifted reversion, implicit equations, tree series, Lambert W , and coefficient forms.	The Lagrange and shifted-reversion chapters, Chapter 28, and the examples in Chapters 33–34.
Catalan_number.txt, Associahedron.txt	Catalan models, Narayana refinements, polygon triangulations, dissections, associahedral faces, and the Tamari order.	The base inversion examples and Chapter 31, where the associahedron index is tied explicitly to its dimension and number of leaves.

37.2 MathWorld dossier

Files or family	Added material	Integrated location
Bell, Bell polynomial, Dobiński, and complementary-Bell files	Signed block weights, the complementary Bell EGF, recurrence, Dobiński-type sum, and convolution inverse of the ordinary Bell sequence.	Chapter 26, with the ordinary Bell theory in the base chapters.
Bernoulli number and polynomial, Bernoulli polynomial of the second kind, modified Bernoulli, Euler polynomial, Genocchi, and Euler–Maclaurin files	A unified Appell-family treatment; reflection, difference, multiplication and power-sum formulas; Genocchi conversion; Cauchy–Gregory coefficients; modified normalization; Euler–Maclaurin with a periodic remainder.	Chapter 29. The symbol β_n prevents collision with Bell numbers.
Pochhammer, digamma, polygamma, and Barnes G files	Rising-factorial parameter derivatives, logarithmic Bell-polynomial formulas, polygamma sums and multiplication, the Weierstrass product for G , Alexeiewsky’s integral, and its asymptotic expansion.	Chapter 30; constants and branch conventions are matched to the NIST normalization [16].

Continued on the next page

Files or family	Added material	Integrated location
Catalan-number files	Narayana numbers, explicit Catalan bijections, polygon dissections, associahedral face counts, Tamari order, and Raney numbers.	Chapter 31.
Darboux-formula files	The finite Darboux identity, its Taylor-remainder proof, algebraic and logarithmic singularity transfer, and the derivation of Catalan asymptotics.	Chapter 32, aligned with the transfer framework of analytic combinatorics [7].
First- and second-kind Stirling files and notebooks	Low-order tables, explicit sums, recurrences, and normalization checks.	The main Stirling parts and Chapter 35. Notebook output was used only as an independent low-order check; the proofs do not depend on a computer algebra system.

The paired PDF and notebook files in the MathWorld directory serve different purposes: the PDFs preserve the human-readable reference pages, while the notebooks provide exact symbolic expressions suitable for spot checks. Neither representation is treated as a proof certificate by itself.

Chapter 38

Formula-status ledger and proof audit

38.1 Classes of source statements

Every substantive source display was assigned to one of four classes before the merger.

1. *Exact identities.* These are stated in normalized notation and proved by a bijection, coefficient extraction, recurrence uniqueness, finite differences, incidence inversion, formal residues, or analytic estimates.
2. *Equivalent coordinate forms.*
Examples include the partition, frequency-vector, and Bell polynomial forms of Faà di Bruno; the matrix, basis-change, and generating-function forms of Stirling inversion; and the coefficient, derivative, and residue forms of Lagrange inversion. They are presented as one theorem with explicit conversions.
3. *Corrected formulas.* An index shift, omitted range, absent Jacobian, ordinary/exponential factorial mismatch, or unstated branch hypothesis is repaired before proof. The type-*B* Eulerian convention, associahedron indexing, multivariate Good determinant, and delta-function Jacobian are the principal examples.
4. *Schematic or computational claims.* Undefined asymptotic placeholders are replaced by recursively defined coefficients. Finite searches and numerical bounds are accompanied by exact recurrences, directed intervals, or modular certificates.

38.2 Reusable proof engines

The long list of formulas reduces to a small number of mechanisms.

Change of polynomial basis

Falling and rising factorials diagonalize finite differences; Stirling and Lah numbers are transition matrices. Eulerian numbers provide the parallel transition between powers and shifted binomial coefficients.

Labelled decomposition

Exponential generating functions turn disjoint labelled components into products and sets of components into exponentials. Bell polynomials retain the complete block-size profile.

Incidence inversion

Möbius inversion on the Boolean lattice gives binomial inversion; on the partition lattice it gives moments versus cumulants. Sheffer and Riordan arrays package the same substitutions as lower-triangular operators.

Formal residues The residue change-of-variables rule proves one-variable Lagrange–Bürmann

and, with a Jacobian determinant, Good’s multivariate formula.

Singularity localization

Cauchy’s formula, saddle points, Darboux’s identity, and transfer theorems convert local analytic structure into coefficient asymptotics. The assumptions needed for uniformity and error terms are stated at the theorem where they are used.

38.3 Independent verification protocol

The exact algorithms in Chapter 35 provide a second layer of protection against transcription errors. A new identity should be checked by:

1. converting every series to one declared ordinary or exponential normalization;
2. expanding far enough to encounter nontrivial multiplicities—fourth order already tests the Faà di Bruno pattern 1, 4, 3, 6, 1;
3. verifying triangular recurrences and inverse-matrix products over exact integers or rationals;
4. using a degree-bounded grid or a size-bounded Chinese-remainder certificate when the symbolic expression is too large for direct comparison; and
5. keeping analytic remainder estimates separate from finite numerical checks.

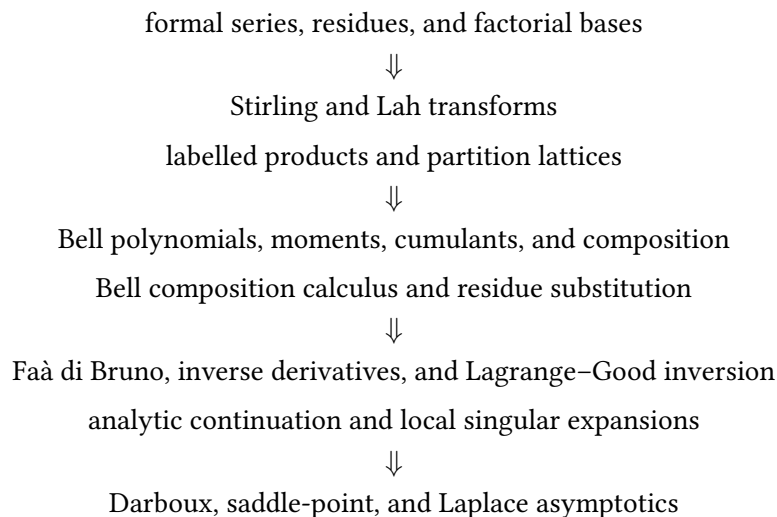
This protocol does not replace derivation. It makes the derivation reproducible and catches the factorial, sign, and index errors most likely to survive visual review.

38.4 External cross-checks

The special-function normalizations and principal identities were cross-checked against the current NIST Digital Library of Mathematical Functions [16]. The multivariate inversion statement and its asymptotic role were checked against Bender and Richmond [2]; the one-variable inversion framework was compared with Gessel’s proof survey [8]. These references are used for provenance and normalization. The proofs in the body are self-contained at the level of formal coefficient calculus developed here.

38.5 Dependency map

The mathematical dependence of the unified article is summarized by



Eulerian and Catalan families form two transverse strands: the first translates powers into descent and polyhedral data, while the second translates reversion into trees, paths, polygon dissections, and associahedra. The resulting structure is a coefficient calculus rather than a catalogue of isolated identities.

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- [24] The supplied `mathworld/` dossier: PDF and notebook reference copies on Bell, Bernoulli–Euler–Genocchi, Catalan, Darboux, Pochhammer, polygamma, Barnes G , Dobinski, and Stirling topics.
- [25] The five LaTeX drafts in the supplied `articles/` directory, merged and normalized for this project. The draft-by-draft contribution record appears in Chapter 36.

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